Part II

Options: Mechanics and Strategies

Three chapters that take an option apart without pricing it. The mechanics come first (how contracts are listed, adjusted, and exercised), then the arbitrage bounds that hold whatever model one believes, and finally the standard strategies. Chapter 11 deserves more attention than its lack of a model suggests: those bounds are the constraints every later model must respect, and they follow from no-arbitrage alone.

10 **Mechanics of Options Markets**
listing, adjustment, exercise, and the life cycle of a contract

11 **Properties of Options and Arbitrage Bounds**
put–call parity, and the limits every model must respect

12 **Trading Strategies Involving Options**
spreads, combinations, and the profit diagrams that describe them

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Chapter 10
Mechanics of Options Markets

Apart from the delivery options embedded in Treasury futures (Section 6.4), everything in Part I was symmetric. A forward, a futures contract, and a swap all oblige both parties, and both can gain or lose without limit. This chapter begins the study of instruments that are not symmetric, and the asymmetry changes almost everything: how they are priced, how they are margined, who can safely write them, and what happens at expiration.

The economics were sketched in Section 1.1. An option confers a right rather than an obligation, so its holder will exercise only when it pays; the payoff can therefore never be negative to the holder, and they must pay for it at the outset. What Part III must supply is the price of that right. First, though, comes the machinery: what exactly the contract says, how it is quoted, who guarantees it, what the writer must post, and what happens at expiration. Getting this wrong is a common source of loss, and unlike model error it is avoidable.

Sections 10.1 to 10.4 are the contract and how it is read: the four positions, the specifications, the underlyings beyond single stocks, and the option chain. Sections 10.5 and 10.6 are how options trade and are quoted, including the volatility quotation Chapter 15 explains. Sections 10.7 to 10.9 are the mechanics of short selling, margin, and exercise and assignment; Section 10.10 looks beyond the standard contract and Section 10.11 at who trades options and why.

This chapter also covers a development with no counterpart in earlier textbooks: options markets have been transformed since 2020 by retail participation and by contracts that expire the same day they are traded. Section 10.12 treats it as a market-structure topic rather than a curiosity, because it now shapes intraday behavior in the largest equity index in the world.

10.1 CALLS, PUTS, AND THE FOUR POSITIONS

A **call** gives its holder the right to buy the underlying asset at the **strike price** K; a **put** gives the right to sell at K. The right lasts until the **expiration date**. A **European** option can be exercised only at expiration; an **American** option can be exercised at any time up to it. The names, coined by Paul Samuelson in the mid-1960s, have nothing to do with geography: both types trade on both continents.

Every option contract has two sides. The holder is said to be **long** and has **bought** the option; the writer is **short** and has **sold** it. There are four elementary positions, shown in Figure 10.1. Like Figure 1.1, the figure plots profit, the payoff net of the premium, because that is what a trader experiences.

The two right-hand panels are the ones to dwell on. A writer's best case is the premium, achieved when the option expires worthless, and their worst case is unbounded for a call and bounded only by the stock reaching zero for a put. This is why the two sides of an option trade are not mirror images in any practical sense: they face different margin requirements, different capital treatment, and different credit risk. The buyer of an option has no obligations after paying the premium and can be a retail client with a small account. The writer has an open-ended obligation and must be collateralized, which is the subject of Section 10.8.

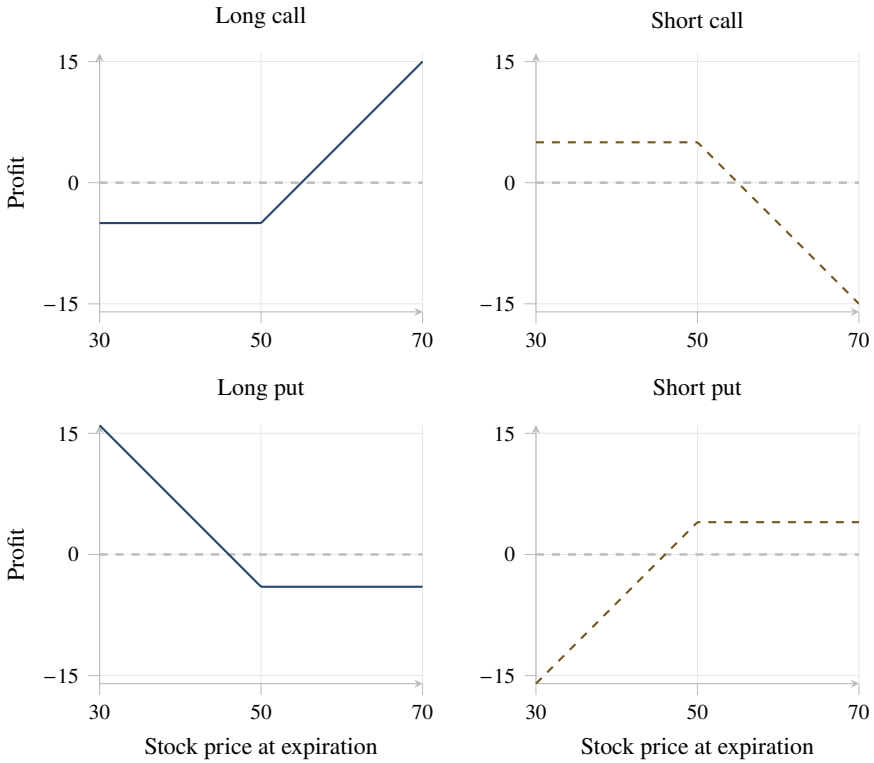


Figure 10.1: Profit at expiration. Strike 50; call premium 5, put premium 4.

Moneyness. An option's **moneyness** is where the stock price S stands relative to the strike K. An option is **in the money** if exercising it immediately would produce a positive cash flow, **at the money** if the stock price equals the strike, and **out of the money** otherwise. For a call these correspond to $S > K$, $S = K$, and $S < K$; for a put the inequalities reverse. The terms describe the option, not the trade: an out-of-the-money option can be an excellent purchase and an in-the-money one a poor purchase, depending entirely on price.

Intrinsic and time value. The **intrinsic value** of a call is $\max(S - K, 0)$, and of a put $\max(K - S, 0)$: what the option would be worth if it expired now. An American option must trade for at least its intrinsic value, since otherwise one could buy it and exercise immediately for a riskless profit. The difference between the market price and the intrinsic value is the **time value**, and it exists because the remaining life gives the stock further opportunity to move in the holder's favor while the downside stays capped.

Figure 10.2 shows the relationship that Part III spends several chapters making precise. Suppose the stock is at 50 and a call struck at 50 trades at 5. Its intrinsic value is zero, so all 5 is time value. With the stock at 60 the curve puts the call at about 12: 10 of intrinsic value and about 2 of time value. With the stock at 40 it is worth about 2, again all time value. Two features generalize. Time value is greatest near the money, because that is where uncertainty about whether the option finishes in or out of the money is greatest; deep in-the-money and out-of-the-money options have little of it. And time value decays to zero at expiration, when the curve collapses onto the kink, the phenomenon Chapter 18 quantifies as theta.

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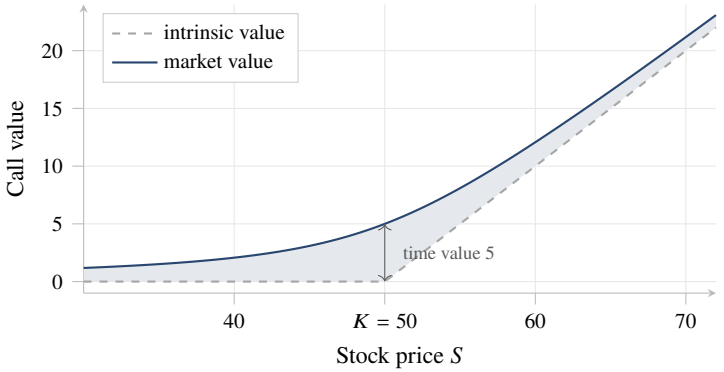


Figure 10.2: A call before expiration, assumed to trade at 5 with the stock at the strike of 50. Shaded region is time value. Schematic.

10.2 CONTRACT SPECIFICATIONS

An exchange-traded option is a standardized contract, and the standard terms matter because a trader chooses among them rather than negotiating them.

Contract size. A standard US equity option covers 100 shares. A quoted premium of $3.20 therefore costs $320 per contract, and a novice who confuses the two has misjudged their position by two orders of magnitude.

Strike prices. The exchange lists strikes at fixed intervals around the current price and adds new ones as the stock moves. The base rule for US equity options sets the interval by price level: $2.50 for strikes up to $25, $5 from $25 to $200, and $10 above $200. Listing programs have narrowed that considerably. The $2.50 Strike Price Program extends the $2.50 interval on selected classes to $50, and to $100 within $10 of the stock price; most actively traded stocks carry $1 intervals below $50, and the lowest-priced stocks $0.50; the largest ETFs and indices are denser still: SPY, the largest S&P 500 ETF, is listed at $1 intervals whatever its price, and S&P 500 index options at 5-point intervals near the money, widening to 25 and 50 points farther out. An actively traded $48 stock therefore shows strikes at 45, 46, 47, and so on up to 50, then 52.50, 55, 57.50, and 60. New strikes are added as the stock nears either end of the ladder, so the price stays bracketed, and old ones stay listed until they expire. This is why a stock that has doubled carries a long tail of near-worthless low-strike puts beneath it.

Expiration dates. US equity options historically expired monthly, on the third Friday. That is still the anchor for the largest contracts, but the calendar has filled in dramatically. Weekly expirations were introduced in 2005 and extended across the most liquid stocks, and since the week of January 26, 2026, after SEC approval that month, stocks with a market value above $700 billion and the largest bitcoin ETF have also carried Monday and Wednesday expirations. For the S&P 500 index, expirations now occur *every trading day*, following the addition of Tuesday and Thursday expirations in 2022. Figure 10.3 shows what this looks like over a typical month, and Section 10.12 explains why it matters.

Exercise style. Options on individual US stocks are American; options on broad indices are typically European. The distinction is not cosmetic. Chapter 11 shows that an American call on a non-dividend-paying stock should never be exercised early, that an American put may well be, and that the difference in value between the two styles can be substantial for puts.

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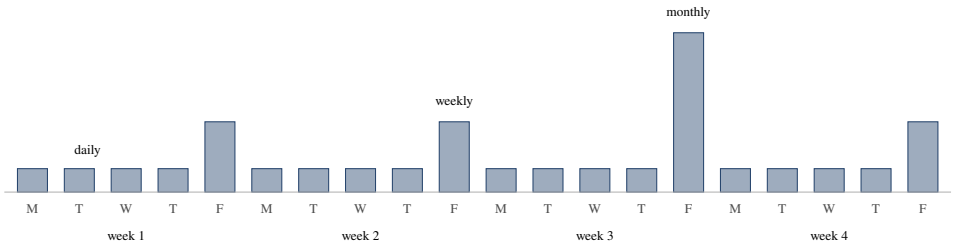


Figure 10.3: S&P 500 index option expirations in a month. Bar heights schematic.

Settlement. Equity options deliver shares. Index options settle in cash against a value computed from opening or closing prices, because delivering a 500-stock basket is impractical.

Corporate actions and contract adjustment

An option is a claim on a specified quantity of a specified security. When a corporate action changes what a share *is*, the contract must be adjusted, or the option would be arbitrarily enriched or destroyed by an event unrelated to the company's value.

The governing principle is to preserve the aggregate value of the position. For a stock split or stock dividend, the contract multiplier and strike are adjusted so that the total strike cost is unchanged. For a merger, the deliverable becomes whatever a holder of 100 shares receives. Ordinary cash dividends, paid under a regular policy, do *not* trigger an adjustment, however large; a special dividend of \$0.125 a share or more does, normally by reducing the strike. This is important and often misunderstood, and Chapter 11 shows that it is precisely why dividends can make early exercise of American calls rational.

Example 10.1 A three-for-two split

An investor holds 10 call contracts on a stock, strike \$60, each covering 100 shares. The company declares a 3-for-2 stock split; each old share becomes 1.5 new shares, and the stock price falls from about \$66 to about \$44.

Without adjustment the options would be destroyed: a \$60 strike on a \$44 stock is deep out of the money, though nothing of economic substance has happened. The adjustment multiplies the deliverable by $3/2$ and divides the strike by the same factor:

$$\text{new contract size} = 100 \times \frac{3}{2} = 150 \text{ shares}, \qquad \text{new strike} = \frac{60}{3/2} = \$40$$

Check that the position is unchanged. Before, exercising cost $10 \times 100 \times 60 = \$60{,}000$ for 1,000 shares worth \$66,000. After, exercising costs $10 \times 150 \times 40 = \$60{,}000$ for 1,500 shares worth $1{,}500 \times 44 = \$66{,}000$. The strike cost, the share value, and the moneyness are all preserved.

The number of contracts stayed at 10 and the contract size changed. For splits producing whole-number ratios (2-for-1, 3-for-1) the convention runs the other way: the contract count is multiplied and the size stays at 100. Either way the aggregate is preserved, but a trader reading a position report must know which convention was applied.

How an option is named

Every listed option needs an identifier, and unlike a stock ticker it must encode four things at once: the underlying, the expiration, the type, and the strike. Since 2010 the US market has used the format established by the Options Symbology Initiative, shown in Figure 10.4.

The symbol in the figure reads: Apple, expiring January 19, 2024, call, strike \$190.000. Nothing has to be looked up: the identifier is the contract description.

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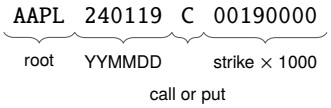


Figure 10.4: US option symbol format.

The strike is carried in eight digits with three implied decimals, so `00190000` is \$190.000 and `00002500` is \$2.50. The three decimal places exist precisely because adjustments of the kind in Example 10.1 can produce strikes that are not round numbers: dividing \$60 by $3/2$ gave a clean \$40, but dividing by $7/5$ would not have.

This scheme replaced a much cruder one. Before 2010, an option was identified by a root plus two letters, one encoding the expiration month and type and one encoding the strike. With only twenty-six letters available for strikes, the system could not survive the proliferation of strike prices and expirations described earlier in this section, and by the late 2000s it was producing genuine ambiguity about which contract a symbol referred to.

The adjusted-contract trap. One convention deserves emphasis because it causes real losses. When a contract is adjusted for a corporate action, it does not keep the ordinary root but receives a *numbered* one: `AAPL1`, then `AAPL2` for a second adjustment, and so on. The digit is appended whatever the length of the root; the root field of the symbol is six characters wide, so even a five-letter root such as `GOOGL` has room for it. The two roots then trade side by side, and they are different contracts. `AAPL` delivers 100 shares; `AAPL1` may deliver 150 shares, or shares plus cash, or shares in an acquiring company.

A trader who sees an unusually cheap quote on what appears to be a familiar option is very often looking at an adjusted contract with a different deliverable, and a position report that lists both roots under the same underlying will misstate the exposure. The rule is simple: a numbered root means read the adjustment memorandum before trading.

10.3 OPTIONS ON INDICES, ETFS, CURRENCIES, AND FUTURES

Options on individual shares are the natural teaching case but not the largest market. Table 10.1 sets out how the specifications differ across underlyings, because the differences change what a position actually is.

Table 10.1: Specifications across underlyings, US market.

Underlying	Size	Exercise	Settlement	Note
Single stock	100 shares	American	Deliver shares	Dividends drive early exercise
Stock index	Multiplier × index	European	Cash	Settlement value may be untradeable
Index ETF	100 shares	American	Deliver shares	Index economics, stock mechanics
Currency	Fixed currency amount	European	Cash or delivery	Foreign rate acts like a dividend yield
Futures	One futures contract	American	Futures position	Exercise yields a margined position

Index options. A stock index option has a multiplier (\$100 for the main US index contracts), so a contract on an index at 5,250 controls $5{,}250 \times 100 = \$525{,}000$ of notional. These are

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the largest option contracts most participants will encounter. Example 10.2 works through a position.

Cash settlement introduces a subtlety. Monthly index options settle against a value computed from the *opening* prices of the constituent stocks, which do not all open at once. The resulting settlement value can differ from any index level published during the session, and a holder cannot trade at it. A position carried into settlement is therefore exposed to a price that is calculated rather than observed.

Example 10.2 An index option position

An index stands at 5,250. A fund buys 10 call contracts struck at 5,300, expiring in one month, quoted at 42.00. The multiplier is \$100.

The position costs $10 \times 100 \times 42.00 = \$42{,}000$ and controls $10 \times 100 \times 5{,}250 = \$5{,}250{,}000$ of index.

The fund has obtained exposure to \$5.25 million of index for \$42,000 (eight-tenths of one percent) and, unlike the futures position of Example 1.5, it can never be asked for another cent. That is the main argument for using options when funding is uncertain.

If the index settles at 5,380, the payoff is $(5{,}380 - 5{,}300) \times 100 \times 10 = \$80{,}000$ and the profit is \$38,000. If it settles anywhere at or below 5,300 the options expire worthless and \$42,000 is lost. The index must rise about 1% merely to reach the strike, and about 1.8% to break even: buying out-of-the-money options requires being right about direction, size, and timing simultaneously, which is why Chapter 12 builds structures that relax one of those requirements.

ETF options. Options on an index-tracking ETF give near-identical economic exposure but with the mechanics of a stock option: American exercise, delivery of ETF shares, and dividends paid by the fund. Since November 2024 they have included the spot bitcoin ETFs, the largest now with a position limit of 1,000,000 contracts . A trader who is short ETF calls can be assigned early (made to deliver because a holder has exercised, the mechanism of Section 10.9) around an **ex-dividend date**, the first day the shares trade without the coming dividend, because a holder can exercise the day before to collect it; a trader short European index calls such as SPX cannot be assigned at all before expiration. (The S&P 100 trades both ways: OEX, the first index option, listed by the CBOE in 1983, is American, and XEO is its European twin.) The two positions look interchangeable on a payoff diagram and are not interchangeable in a risk system.

There is also a tax distinction, and in the US it is large enough to drive the choice between the two. Broad-based index options are "Section 1256" contracts: every gain is split, with 60% taxed at the long-term capital gains rate and 40% at the short-term rate, however briefly the position was held, and open positions are marked to market at year end. Options on an index ETF are ordinary equity options and receive no such treatment: a position held for a month is taxed entirely at the short-term rate. For a taxable US investor trading short-dated index exposure, this is often a larger consideration than the bid–ask spread.

Currency options. A currency option is written on a fixed amount of foreign currency, and its valuation treats the foreign interest rate exactly as a dividend yield, the reasoning of Section 5.7 carried into option pricing. Chapter 16 develops it.

Options on futures. Exercising an option on a futures contract does not deliver an asset; it delivers a *position* in the futures contract, marked to market from the start. Exercising a futures call therefore produces a long futures position at the most recent settlement price plus a cash payment of that settlement price less K; a holder who closes the position at once collects

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the intrinsic value at the futures price then prevailing. Because the resulting position is itself margined, an option on futures never requires the holder to fund the underlying, which is why these contracts dominate exchange-traded options on commodities and interest rates, where producers and processors use them to set a floor or a ceiling on what they sell or buy, and banks and asset managers to hedge rate moves. Chapter 16 prices them with Black's model, and Chapter 6's SOFR and Treasury contracts are the underlyings that matter most.

10.4 READING AN OPTION CHAIN

Options are quoted in a grid (strikes down, expirations across) known as the **option chain**. Table 10.2 shows one expiration for a stock at \$48.30. The chain is shown at \$2.50 intervals for compactness; an active stock at this price would also carry the \$1 strikes between them.

Table 10.2: Calls and puts on a stock at \$48.30, 45 days to expiration. Vol is volume and OI open interest.

Calls				Strike	Puts			
Bid	Ask	Vol	OI		Bid	Ask	Vol	OI
6.55	6.75	120	3,104	42.50	1.55	1.62	890	9,220
4.35	4.50	410	5,870	45.00	1.81	1.89	2,140	14,505
2.62	2.72	3,880	21,440	47.50	2.52	2.63	3,410	18,760
1.44	1.51	5,260	28,115	50.00	3.80	3.95	1,505	11,230
0.68	0.74	2,940	19,680	52.50	5.55	5.75	240	4,010
0.28	0.33	1,110	12,450	55.00	7.60	7.90	60	1,845

Example 10.3 What the chain is telling you

Read Table 10.2 carefully and several facts emerge that no formula supplied.

Where the trading is. Volume and open interest peak at the 50 and 47.50 strikes, just above and below the \$48.30 spot. Liquidity concentrates near the money, and a trader wanting the 55-strike call must accept a bid–ask spread that is wider in proportion to the premium.

The cost of the spread. The 50-strike call is 1.44 bid, 1.51 ask. A trader who buys and immediately sells loses \$0.07 per share, or \$7 per contract, or 4.7% of the position's value. On the 55-strike call the spread is \$0.05 on a \$0.30 option, which is over 16%. Bid–ask spreads on cheap options are enormous in percentage terms, and a strategy that trades far out-of-the-money options frequently can be defeated by transaction costs alone, regardless of whether the view is right.

Volume versus open interest. Volume is contracts traded today; open interest is contracts outstanding. At the 42.50 put, volume of 890 against open interest of 9,220 says most holders are sitting still. High volume relative to open interest (the 47.50 call, 3,880 against 21,440) indicates active repositioning.

An internal check. Put–call parity, derived in Chapter 11, requires that a call and put at the same strike be linked through the stock price and the discounted strike. At the 47.50 strike, the call's midpoint, 2.67, and the put's, 2.575, differ by \$0.095. With no dividends and a 5% rate over 45 days, the difference would have to be

$$S - Ke^{-rT} = 48.30 - 47.50\,e^{-0.05\times 45/365} = 48.30 - 47.208 = \$1.092$$

The shortfall of about \$1.00 is mostly the market's expectation of dividends payable before expiration, together with any cost of borrowing the stock, which the quotes alone cannot separate from them. Every strike in the chain implies about the same \$1.00, so the chain is internally consistent, and Section 11.6 shows how to read the number off the quotes rather than guessing at it.

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10.5 TRADING AND MARKET MAKING

US listed options trade on more than a dozen exchanges, all clearing to the same clearing house, with orders routed among them by price and by economics that the end investor does not see. The SEC regulates them and FINRA the brokers that sell them; options on futures are futures products, regulated by the CFTC. Two features of this structure have direct consequences.

Market makers quote a bid and an ask continuously and are the counterparty to most retail trades. They do not take directional views; they earn the spread and hedge the resulting exposure, dynamically, using the underlying and other options. Chapter 18 is about how that hedging works. What matters here is the consequence: a market maker who has sold calls is short the option and must buy stock as the stock rises, which means dealer hedging *transmits* option activity into the underlying market. Section 10.12 shows what happens when the option activity becomes large enough relative to the stock.

Payment for order flow. Retail brokers in the US route customer orders to wholesalers (large market makers that fill retail orders), which pay for the privilege, and since 2019 this has funded zero-commission trading. The trade is not free; the cost has moved from an explicit commission into the execution price, and it is the harder cost to see. Whether retail investors are better off on net is contested: execution quality improved for stocks, but the picture is murkier for options, where spreads are wider and **price improvement** (the amount by which a fill beats the best displayed quote) is measured against a quote so wide that beating it says little.

Position limits, order types, and legging risk

Three further pieces of trading mechanics have practical consequences.

Position and exercise limits. Exchanges cap the number of contracts any one participant may hold on the same side of the market (calls held plus puts written, or puts held plus calls written), with limits ranging from tens of thousands of contracts on less actively traded stocks to 250,000 on the most liquid stocks, and millions on the largest ETFs. Exercise limits cap how many may be exercised within any five consecutive business days, generally at the same number as the position limit. The purpose is to prevent the option market being used to corner the underlying: a call position large relative to a stock's free float (the shares actually available for trading), exercised at once, can force short sellers to buy back shares at inflated prices. Bona fide hedgers can seek exemptions, as from the futures limits of Section 2.1.

Order types. Because option bid–ask spreads are wide relative to the premium (Example 10.3 showed 4.7% and 16%), a market order is genuinely dangerous. It executes at the ask when buying and the bid when selling, on a quote that may itself have widened in the instant before it arrives. Professional practice is to place limit orders at or near the midpoint and accept non-execution; a retail trader who submits market orders in options pays the full spread on every round trip.

Legging risk. A strategy with two or more option legs can be executed leg by leg or as a package. Executing separately exposes the trader to the market moving between fills, which on a two-option position worth \$0.40, built from options worth \$2.60 and \$2.20, can easily exceed the value of the position itself. Exchanges therefore maintain complex order books that match multi-leg orders as a single package at a net price. Chapter 12 builds strategies from several legs throughout, and every one of them should be understood as a package rather than as a sequence of trades.

10.6 QUOTING IN VOLATILITY

A feature of professional option markets surprises newcomers: dealers do not negotiate the price. They negotiate the implied volatility, which Section 15.7 defines as the volatility that, put into the pricing formula, returns the option's price. The price follows once the trade is agreed.

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The reason is practical. An option's price depends on the underlying price, the strike, the time remaining, interest rates, and volatility. Four of those five move continuously or are fixed by the contract; only volatility expresses a view. If a dealer quotes "$1.47" and the stock moves a penny before the client responds, the quote is stale. If the dealer quotes "21.5 volatility," the quote survives the stock moving, and the price is struck at the end from the agreed volatility and an agreed reference spot.

Two consequences follow, and both matter for Part III.

First, large OTC option trades are commonly done with a **delta exchange**: the option trades together with an agreed quantity of the underlying, sized to the option's directional exposure (its delta, in the language of Chapter 18), so that neither side takes on a position in the stock they did not want. The trade is a pure volatility transaction, which is what both sides intended.

Second, and more fundamentally: implied volatility is a price, not a model output. It is tempting to regard it as something extracted from an option price by inverting a formula. In the professional market the causation runs the other way: volatility is quoted, negotiated, and traded, and the dollar price is derived from it. Chapter 19 studies the resulting surface of quoted volatilities across strikes and maturities as a market object in its own right, and this is why.

10.7 SHORT SELLING, BORROW COSTS, AND PUTS

One more piece of mechanics connects the option market to Part I, and it explains a pattern that otherwise looks like mispricing.

Buying a put and writing a call at the same strike and expiration creates a position that gains and loses exactly as a short stock position does, a **synthetic short**. Chapter 11 derives this from put–call parity. It means an investor who wants short exposure has two routes: borrow the stock and sell it, or build the synthetic in the option market.

The two routes must cost about the same, which has a direct consequence. Section 5.2 showed that when a stock is hard to borrow and its holders will not or cannot sell it, the no-arbitrage relation degrades from an equality into a band whose width is set by the borrow fee. That band appears in the option market too: on a hard-to-borrow stock, puts trade above the price that put–call parity with the calls implies, by roughly the present value of the borrow cost. Example 10.4 puts a number on it.

Example 10.4 What a borrow fee does to the option chain

A stock trades at $50 and can be borrowed only at a fee of 3% a year. Consider six-month options struck at $50.

An arbitrageur who saw the put priced above what put–call parity implies would sell it, buy the call, and short the stock (the reversal of Chapter 11), but shorting requires borrowing at 3%, which over six months costs

$$50\left(1 - e^{-0.03\times 0.5}\right) = 50 \times 0.014888 = \$0.74$$

per share. The trade is therefore unprofitable unless the put is more than 74 cents rich, and the option market can sit anywhere inside that band without offering a profit to anyone who must borrow the stock. (A holder willing to sell shares already owned avoids the fee, which is why the band appears only where such holders are scarce, as they usually are on a heavily shorted stock.)

So a chain showing puts 60 cents rich to naive put–call parity is not mispriced and does not indicate bearish sentiment. It is quoting the borrow fee. A trader who reads it as a signal has read a financing cost as an opinion. Chapter 19 finds something similar in the volatility skew, the pattern in which low-strike options trade at higher implied volatilities than high-strike ones: part of it is a price set by one-sided demand for protection rather than a forecast.

The magnitude scales with the fee, and for genuinely hard-to-borrow stocks the fee can reach tens of percent, at which point the put becomes the only practical way to be short and its price says

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almost nothing about the expected direction of the stock.

10.8 MARGIN REQUIREMENTS

The asymmetry of Figure 10.1 produces an asymmetry in collateral, and it is the cleanest illustration in the book of margin following risk rather than notional.

The buyer posts nothing. A purchased option is paid for in full at inception (US rules let a listed option with more than nine months to run be bought with 75% paid and the rest borrowed) and creates no further obligation. The maximum loss is the premium, already paid. This is what makes options valuable to a hedger with limited access to liquidity (Section 12.10). (The exception is an option margined futures-style, Section 16.5, whose value change is settled daily.)

The writer posts margin. A **naked** call writer, one who has sold calls without owning the shares, has unlimited exposure and must collateralize it. The standard US requirement is the greater of two formulas:

$$\text{(a)}\quad 20\%\text{ of underlying value} - \text{out-of-the-money amount} + \text{option value} \tag{10.1}$$

$$\text{(b)}\quad 10\%\text{ of underlying value} + \text{option value} \tag{10.2}$$

The option value is the premium on the day of the trade and the market price thereafter, so the requirement is recomputed daily as the option moves. The second formula is a floor that binds for far out-of-the-money options, where the first would permit a very large position on very little collateral. A naked put writer posts the greater of the same two formulas, with the out-of-the-money amount measured as how far the stock stands above the strike and the floor in (b) taken as 10% of the strike price rather than of the underlying value. Example 10.5 applies both to a call.

Example 10.5 Margin on a naked call

A trader writes 10 naked call contracts, strike \$55, on a stock at \$48.30, collecting \$0.30, about the midpoint of the 0.28–0.33 quote in Table 10.2. The premium received is $10 \times 100 \times 0.30 = \$300$.

Underlying value is $10 \times 100 \times 48.30 = \$48{,}300$, and the options are \$6.70 out of the money, or $10 \times 100 \times 6.70 = \$6{,}700$ in aggregate. From (10.1),

$$0.20 \times 48{,}300 - 6{,}700 + 300 = 9{,}660 - 6{,}700 + 300 = \$3{,}260$$

and from (10.2),

$$0.10 \times 48{,}300 + 300 = \$5{,}130$$

The requirement is the greater, \$5,130.

The trader has collected \$300 and must post \$5,130, seventeen times the premium, to support a position whose maximum gain is that \$300 and whose maximum loss is unbounded. If the stock rallies to \$70, the calls are worth \$15 and the position has lost $10 \times 100 \times 15 - 300 = \$14{,}700$, nearly three times the margin originally posted. As in Example 1.5, *margin is a deposit and not a maximum loss*, and the writer will be called for the difference.

This is the arithmetic behind a standing piece of market wisdom: selling far out-of-the-money options produces a long run of small gains punctuated by losses that erase them. The strategy requires capital, discipline, and the ability to survive the tail. It is not a yield-enhancement technique, whatever it is marketed as.

Covered positions require less. A call writer who owns the underlying shares has no unbounded exposure (the shares are the deliverable) and posts no additional margin beyond the stock itself. A cash-secured put writer who holds K per share in cash is similarly covered. These structures dominate retail option writing for exactly this reason.

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Portfolio margining evaluates the whole position rather than each leg, stressing the portfolio across a grid of underlying prices and volatilities and requiring collateral for the worst outcome. The methodology is the option-market cousin of the portfolio margining of Section 2.3 and of the initial margin models of Chapter 8. It substantially reduces requirements for genuinely hedged positions and, as Section 5.14 illustrated with Amaranth, thereby permits positions whose size becomes the risk.

10.9 THE CLEARING HOUSE, EXERCISE, AND ASSIGNMENT

Every US listed option is cleared by the Options Clearing Corporation (OCC), which was established in 1973 alongside the Chicago Board Options Exchange (CBOE) and remains the sole clearing house for the market. The OCC becomes buyer to every seller and seller to every buyer, so an option holder never depends on the solvency of the person who wrote it. The mechanism and the risk it concentrates are those of Section 2.4.

Exercise. Most positions are never exercised but closed by an offsetting trade, the holder selling the option and the writer buying it back, which also captures any remaining time value. A holder exercising an American option notifies their broker, who notifies the OCC. At expiration, options that are in the money by one cent or more are exercised *automatically* ("exercise by exception") unless the holder instructs otherwise. This default prevents value being abandoned through inattention, but it does not apply to an option that finishes exactly at the strike, the trap of Example 10.6.

Assignment. The OCC does not know which particular writer should be assigned, and it does not try to. It allocates exercise notices *randomly* among clearing members with short positions, who in turn allocate randomly or first-in-first-out among their clients. A writer therefore cannot know in advance whether they will be assigned, only estimate the probability. A writer of an in-the-money American option can have it exercised against them at any moment.

Example 10.6 Pin risk at expiration

A market maker is short 200 put contracts struck at \$50 and has hedged by shorting 10,000 shares, roughly the hedge Chapter 18 will call the delta hedge: with the stock near \$50 the puts behave like about half of their 20,000 shares. As expiration approaches, though, the puts are almost certain to be exercised if the stock finishes below \$50 and certain to expire worthless if it finishes above.

The stock closes at exactly \$50.00 on expiration Friday.

If every holder exercises, the writer is assigned 20,000 shares at \$50, which covers the 10,000-share short and leaves 10,000 shares to sell. If no one exercises, the writer is left short 10,000 shares they did not want. In practice some holders exercise and some do not, for reasons the writer cannot observe, and the writer learns the outcome only after the market has closed: on Saturday morning, with the next opportunity to trade on Monday.

The exposure is therefore an unhedged overnight-plus-weekend position of unknown size and unknown direction, on up to 10,000 shares. This is **pin risk**, and it is why traders close positions at strikes the stock is likely to finish near, rather than carrying them into expiration for the last few cents of premium.

A related effect runs the other way. A dealer who is *long* options struck near the spot (long gamma, in Chapter 18's terms) hedges by buying stock when it falls below the strike and selling when it rises above, which mechanically damps movement around that price. Stocks with large open interest at a round strike tend to gravitate toward it as expiration approaches. This "pinning" effect is a hedging artifact rather than a market view. The writer in this example, being short, would trade the other way, which is why pinning depends on who holds the options at the strike.

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10.10 BEYOND THE STANDARD CONTRACT

Three families of option-like instruments sit outside the listed market and are worth distinguishing, because their differences are exactly the features a reader should be alert to.

Warrants are options issued by a company on its own stock, usually attached to a bond or share issue so that it raises capital more cheaply, while holders gain long-dated upside. Exercise creates new shares, so the payoff is diluted, which changes the valuation (Chapter 17). A listed call, by contrast, is written by another participant and exercise transfers existing shares.

Convertible bonds are debt with an embedded option to convert into equity. The holder owns a bond plus a call; the issuer has sold that call in exchange for a lower coupon. They form a large market in embedded options, and Chapter 25 treats them together with credit.

Employee stock options are non-transferable, forfeited if unvested when the employee leaves, and long-dated. Because holders cannot sell them, they exercise early for liquidity and diversification reasons that no arbitrage-based model captures, which is why the valuation of employee options is a distinct problem, the subject of Chapter 17.

Alongside them sits the over-the-counter market in options proper. These are options in the full sense of Section 10.1, negotiated bilaterally with a dealer rather than listed. The listed market is the visible one; by notional the OTC market is larger, and it is where most currency options, most interest rate options (the caps, floors, and swaptions of Chapter 29), and every exotic structure of Chapter 34 trade. The reason is customization. An exchange lists a grid of strikes and dates on a hundred shares of one stock; a dealer will write any strike and any date, on a basket, with a barrier, on an average price, or paid in another currency, and will do so in an amount that would move a listed market visibly. The terms are documented under the ISDA Master Agreement of Section 1.5, the price is negotiated as a volatility in the manner of Section 10.6, and the buyer is exposed to the dealer's default for the life of the trade unless the agreement is collateralized under a CSA of the kind Chapter 8 describes or the contract is standardized enough to clear. The credit valuation adjustment of Chapter 9 is the price of that difference. The users are the ones the listed market cannot serve: a treasurer hedging a receivable whose date and amount match no listed contract, an asset manager buying a swaption against a pension fund's interest rate exposure, an issuer of the structured notes of Chapter 35 hedging the option it has sold to retail investors. What the buyer gives up is the exchange's transparency: there is no public last trade, the dealer's quote is the only price, and a position is unwound with the same dealer or not at all.

Flexible exchange (FLEX) options, introduced in 1993, are the listed market's answer: exchange-listed, but with terms the parties choose, namely any strike, any expiration, either exercise style. They occupy the middle ground the cleared-OTC market occupies in Section 1.4, offering customization with a clearing house behind it. **Long-term equity anticipation securities (LEAPS)** are simply long-dated listed options, extending up to about three years (five on the S&P 500).

10.11 WHO TRADES OPTIONS, AND WHY

Options serve purposes the linear instruments of Part I cannot, and it is worth naming who is on each side before Part III begins pricing them.

Protective put buyers own an asset and want a floor under it. The put is insurance: a premium is paid, the downside is capped, the upside is retained. Pension funds, insurers, and founders and executives with concentrated stock positions are the natural buyers, and Section 12.10 explains why a futures hedge is not a substitute.

Covered call writers own an asset and sell part of the upside for income. Asset managers run this as a systematic strategy, and it is among the most widely used retail option strategies. It converts an uncertain equity return into a more certain but capped one, and it performs poorly

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in the strong markets that investors hold equities for, as Example 10.7 shows.

Example 10.7 Writing a covered call

An investor owns 500 shares of the stock in Table 10.2, trading at \$48.30, and writes 5 call contracts struck at \$52.50 at the \$0.68 bid, 45 days out. Premium received is $5 \times 100 \times 0.68 = \$340$.

Three outcomes at expiration follow, on the 500-share position:

Stock at \$46. The calls expire worthless. The shares have lost $500 \times 2.30 = \$1{,}150$, offset by the \$340 premium, for a net loss of \$810 against \$1,150 unhedged.

Stock at \$52.50. The calls expire just at the money and worthless. The shares have gained $500 \times 4.20 = \$2{,}100$, plus \$340, for \$2,440. This is the best possible outcome, and every price at or above the strike earns it.

Stock at \$58. The calls are assigned at \$52.50. The result is the same \$2,440, while the unhedged position would have gained $500 \times 9.70 = \$4{,}850$. The investor gave up \$2,410.

The premium is \$0.68 on a \$48.30 stock: 1.4% of downside protection, and the breakeven falls only to \$47.62. The strike is 8.7% above spot, so all upside beyond 8.7% has been sold for that 1.4%.

The maximum return is \$2,440 on \$24,150 of stock, or 10.1% in 45 days, which annualizes to a number that appears prominently in marketing material. Annualizing it is misleading: the maximum cannot be earned repeatedly, because it requires the stock to gain 8.7% or more every 45 days.

The strategy is not unsound: the writer is selling insurance and collecting a premium for it, and Chapter 19 shows that the risk premium is positive on average. But "income" misdescribes the strategy: it is the sale of upside.

Corporate issuers embed options in securities. A callable bond is a bond plus a call sold to the issuer, who pays for the right to refinance with a higher coupon; a convertible is the reverse, a call sold to the holder in exchange for a lower coupon.

Volatility traders take positions in the option without wanting exposure to the underlying at all, hedging the direction away and trading the volatility that remains. Part IV is about this.

Dealers intermediate all of the above and hedge what accumulates. Their aggregate position determines whether dealer hedging damps or amplifies moves in the underlying, the mechanism of Example 10.6 operating at market scale.

Retail investors have become, since 2020, a large enough share of activity to change market structure, which is the subject of the next section.

10.12 THE RETAIL TRANSFORMATION AND SAME-DAY OPTIONS

The US options market more than doubled in volume between 2019 and 2023, and the composition of that growth is as important as its size. Three changes drove it.

The first was the elimination of retail commissions in late 2019, which removed the fixed cost that had made small option trades uneconomic. The second was the arrival, during 2020, of a large cohort of new retail participants trading through mobile applications. The immediate effect was a surge in single-stock call buying, concentrated in short maturities and small orders.

The second-order effect is the interesting one. Retail buying of short-dated calls leaves dealers short those calls and therefore short gamma, in the language Chapter 18 introduces: they are obliged to buy the stock as it rises and sell as it falls, the opposite of the damping behavior in Example 10.6. Chapter 23 develops the mechanism in full. In heavily traded stocks during 2021, this hedging demand was large relative to the free float of the stock, and the feedback from option market to stock market became visible rather than theoretical.

A third change came from the exchange side. Once S&P 500 options expired every trading day, every session contained contracts with hours rather than weeks to run. These are

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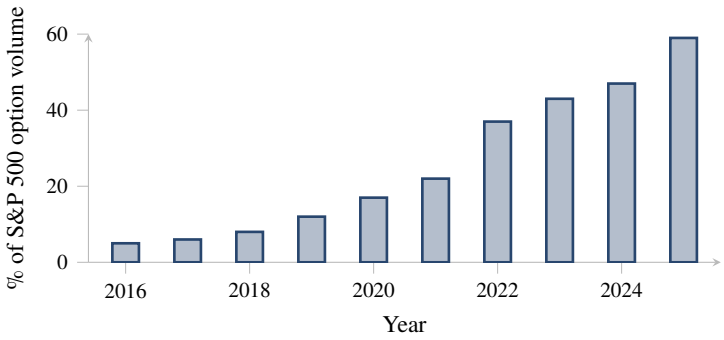


Figure 10.5: Share of S&P 500 index option volume expiring the same day. Approximate.

zero-days-to-expiration (0DTE) options, and Figure 10.5 shows their share of index option volume rising from about 5% to more than half.

Why 0DTE options behave differently. An option with hours to live has almost no time value and enormous gamma, so a dealer hedging a book of them must trade the underlying frequently and in large amounts, carrying option trading into the index during the day. Customers both buy and sell them and the positions close within the day, so dealers carry no large net position overnight; that is why studies so far have found less effect on the index than early commentary feared. Chapter 23 develops the mechanism and weighs that evidence.

Case Study 10.1 What retail bought, and what it cost

The retail options boom is usually discussed in terms of market stability. The more immediate question is what it did to the participants themselves.

Retail option trading is overwhelmingly *buying*, and heavily concentrated in short-dated, out-of-the-money calls: the cheapest contracts on the chain and, as Example 10.3 showed, those with the widest proportional spreads. A \$0.30 option with a \$0.05 spread costs over 16% of its value to enter and exit. A trader turning such a position over weekly faces a transaction cost burden that no directional skill plausibly overcomes.

Layered on top is a structural point from Part III. Buying options is buying volatility, and Chapter 19 documents that implied volatility exceeds subsequent realized volatility on average across most markets and periods. Persistent option buyers therefore pay a risk premium on top of the spread, and should know they are paying it.

None of this argues against retail participation. The cost is simply incurred where no statement shows it: in the spread and in the volatility premium.

SUMMARY

A call confers the right to buy at the strike and a put the right to sell; European options are exercisable only at expiration, American options at any time. A buyer's loss is capped at the premium and the buyer posts no margin; a writer's gain is capped at the premium and the writer's open-ended loss must be collateralized. This asymmetry, not the payoff formula, distinguishes the two sides in practice. An option's value is intrinsic value (what it would be worth exercised now) plus time value, which is largest near the money and decays to zero at expiration.

A standard US equity option covers 100 shares. Contracts are adjusted for splits, mergers, and special dividends to preserve aggregate value, but not for ordinary cash dividends, and an

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adjusted contract trades under a numbered root. The same right differs by underlying: index options settle in cash against a computed value, ETF options can be assigned early around dividends and are taxed differently, and an option on a futures contract delivers a margined position rather than an asset.

Reading an option chain reveals what no model supplies: where liquidity is, how large the proportional bid–ask spread is on cheap options, and whether the quotes are consistent with put–call parity. Professional markets quote options in implied volatility, so a volatility is a traded number and not a model output. On a hard-to-borrow stock the borrow fee appears as puts priced above parity, a financing cost rather than a view.

Naked writers post the greater of two formulas, recomputed daily and far exceeding the premium collected; covered and cash-secured writers post much less, and portfolio margining reduces requirements for hedged positions and thereby enables larger ones.

The OCC clears all US listed options and allocates exercise notices randomly among writers. Options in the money by one cent are exercised automatically at expiration, and positions struck near the closing price create pin risk: an unhedged position of unknown size discovered after the market has closed. The larger over-the-counter market holds most currency and interest rate options and all exotic options, and its buyer is exposed to the dealer's default unless the trade is collateralized or cleared.

Since 2020 zero-commission retail participation and index options expiring every trading day have reshaped the market, and same-day options now account for more than half of S&P 500 option volume. Their hedging carries option trading into the index during the day, and the retail cost lies in bid–ask spreads and the volatility risk premium rather than in commissions.

FURTHER READING

The OCC's *Characteristics and Risks of Standardized Options*, the disclosure document every US options customer receives, states the contract, exercise, and margin rules; the OCC's adjustment memoranda show the principle of Example 10.1 applied case by case, and exchange rulebooks give the current margin formulas. Bryzgalova, Pavlova, and Sikorskaya (2023) is the empirical study of retail option buying and payment for order flow behind Section 10.12's case study. Cboe's research note "The Rise of SPX 0DTE Options" documents the growth that Figure 10.5 plots, and Brogaard, Han, and Won's "Does 0DTE Options Trading Increase Volatility?" is the place to start on the open question Chapter 23 weighs. On payment for order flow, the brokers' Rule 606 routing reports, which disclose what each wholesaler pays for options orders, are the primary data.

PRACTICE QUESTIONS

10.1 An investor buys 5 call contracts at a quoted premium of \$2.85. What is the total cost, and what is the maximum loss?

10.2 Using Table 10.2, compute the proportional bid–ask spread for the 45-strike call and the 55-strike call. Explain why they differ and what it implies for strategy selection.

10.3 A stock trades at \$74 and pays no dividend. A trader writes 8 naked call contracts struck at \$85 for a premium of \$1.10. Compute the margin requirement from both formulas and state which binds.

10.4 A company declares a 2-for-1 stock split. State how the contract terms of an outstanding option are adjusted under each of the two conventions described in Section 10.2, and verify that the aggregate strike cost is preserved in both.

10.5 Explain why an option buyer posts no margin while a writer must, and relate the answer to Figure 10.1.

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10.6 A trader is short 50 call contracts struck at $30 and the stock closes at $30.00 on expiration Friday. Describe the trader's position on Saturday morning and explain what could have been done on Friday to avoid it.

10.7 An option expiring in three hours and one expiring in three months are both at the money. Compare their intrinsic value, time value, and sensitivity to a $1 move in the underlying.

10.8 Distinguish a warrant from a listed call option on the same stock at the same strike. Which is worth more, and why?

10.9 An index option, a currency option, and an option on a futures contract differ in what changes hands on exercise. State what each delivers, and explain why exercising an option on a futures contract leaves the holder with a position to manage rather than an asset to keep.

10.10 Dealers quote many options in volatility rather than in price. Explain what the convention buys them, state what else must be agreed before a volatility quote determines a cash amount, and explain why such a quote stays meaningful while the underlying moves and a price quote does not.

10.11 An investor wants six months of downside protection on a $5 million S&P 500 portfolio, with the index at 5,250 and SPY at $525. Compare buying index puts (SPX) with buying SPY puts: the number of contracts, settlement and exercise, what each leaves the investor holding at expiration, and US tax treatment.

10.12 Name three distinct institution types that buy options and state, for each, the exposure it is buying or removing. For one of them, explain how its demand shows up in the prices the rest of the market sees.

FURTHER QUESTIONS

10.13 Using the 50-strike quotes in Table 10.2, repeat the put–call parity check of Example 10.3 at that strike. Does the chain remain consistent? Identify what you must assume about dividends for the two strikes to agree, and comment on whether that assumption is reasonable.

10.14 Section 10.12 argues that dealer hedging of short-dated retail call buying amplifies moves in the underlying, while Example 10.6 argues that dealer hedging near expiration damps them. Reconcile these. Precisely what determines the sign of the effect, and what would you need to observe to know which regime prevails today?

10.15 Two weeks before the covered call in Example 10.7 expires, the stock is at $55, the calls are quoted $2.50–$2.60, and tomorrow the stock goes ex-dividend with the $1.00 dividend of Example 10.3, after which the calls would be worth about $2.35. Explain why the investor may be assigned tonight, compute what an assignment costs against holding through the ex-dividend date, and state what the investor could have done today to avoid it.

10.16 Contract adjustments preserve aggregate value for splits and mergers but not for ordinary cash dividends. Explain the rationale for treating the two differently, and construct a case where the distinction transfers substantial value from one side of a contract to the other.

10.17 Same-day options do not accumulate an overnight position, which Section 10.12 offers as a reassuring feature. Construct the scenario in which this feature fails to protect the market, and identify what a regulator would need to monitor to see it developing.

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Chapter 11
Properties of Options and Arbitrage Bounds

This chapter contains no model. Nothing in it assumes a distribution for the stock price, a volatility, or a stochastic process, and none of its results will be revised when Chapter 15 supplies a formula. Everything here follows from the absence of arbitrage, and the results are therefore stronger than anything a model can deliver: a model can be wrong, whereas these relationships hold for any model that is not internally inconsistent.

That is worth stating plainly because the temptation is to skip ahead. A reader who knows the Black–Scholes–Merton formula but not put–call parity will price options adequately and misunderstand the market: they will read a price relationship as sentiment, exercise an American call that should have been sold, or construct a "riskless" arbitrage that carries an assignment risk they did not know existed.

Chapter 10 deferred four questions to this chapter: why the option chain in Example 10.3 had to be internally consistent, why a put and a written call together behave like short stock, why dividends make early exercise of an American call rational, and why an American put is worth more than a European one. All four are answered by arbitrage arguments alone.

Section 11.1 lists what determines an option price. Sections 11.2 and 11.3 derive the bounds on a single option and across strikes; Sections 11.4 to 11.7 are put–call parity and what follows from it: synthetic positions, the implied forward, and the box spread as a financing trade. Sections 11.8 to 11.11 are early exercise: why an American call on a stock paying no dividend is never exercised early, why a put may be, what dividends change, and the parity bounds that survive for American options. Section 11.12 asks what an apparent violation actually means, and Section 11.13 who trades these relationships.

11.1 WHAT DETERMINES AN OPTION PRICE

Six quantities determine the value of a stock option: the current stock price S_0, the strike K, the time to expiration T, the volatility σ, the risk-free rate r, and the dividends expected during the option's life. Table 11.1 gives the direction of each effect; the reasoning matters more.

Table 11.1: Effect of increasing one variable, holding the others fixed.

Variable increases	European call	European put	American call	American put
Stock price S_0	+	−	+	−
Strike price K	−	+	−	+
Time to expiration T	?	?	+	+
Volatility σ	+	+	+	+
Risk-free rate r	+	−	+	−
Dividends	−	+	−	+

Figure 11.1 shows the shapes. Two are worth noting beyond their direction, though their

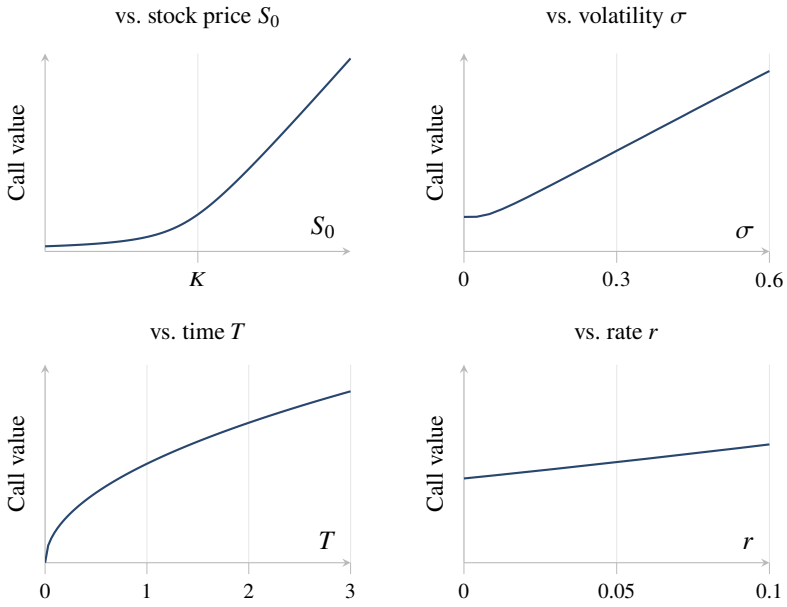


Figure 11.1: A European call, varying one variable at a time. Schematic.

shapes come from the model of Chapter 15 rather than from arbitrage: an at-the-money call's value is flat in volatility at first and then close to linear (the slope is what Chapter 18 names vega, and Chapter 19 relies on the near-linearity when it treats implied volatility as a quoted price), while the dependence on time grows roughly with $\sqrt{T}$, so doubling the maturity does not double the time value.

Stock and strike prices. A call pays $\max(S_T - K, 0)$, which rises with S_T and falls with K.

Volatility. This entry separates options from every instrument in Part I and deserves care. For a forward contract, volatility is irrelevant to the price: the payoff is linear, so increased dispersion raises the good outcomes and lowers the bad ones symmetrically, and they cancel. An option's payoff is not linear. Its downside is truncated at zero, so greater dispersion raises the expected payoff without a matching penalty. *Both* calls and puts increase in value with volatility, which surprises readers who think of a call and a put as opposite bets.

Risk-free rate. Two effects run in the same direction for a call. A higher rate reduces the present value of the strike the holder will eventually pay, and it raises the forward price, $S_0 e^{rT}$ for a stock paying no dividend, on which the call is effectively written. Both increase call values and decrease put values. This is one of the few places where intuition from other markets misleads: rising rates *raise* call prices, holding the stock price fixed.

Dividends. A dividend reduces the stock price on the ex-dividend date without compensating an option holder, who receives no dividends. Calls fall in value, puts rise. Section 11.10 shows this is also what makes early exercise of American calls rational.

Time to expiration, and why two entries are question marks. An American option with more time cannot be worth less, because its holder can do everything the shorter-dated holder can do and more: the longer option's exercise opportunities are a superset.

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European options do not have this property, and the reason is instructive. Consider two European calls on a stock paying a large dividend in one month, one expiring in two weeks and one in six weeks. The shorter option expires before the stock drops on the ex-dividend date; the longer one does not. The shorter option can be worth more. A European put can lose value with more time for a different reason when rates are high and the put is deep in the money: the extra time defers the receipt of K. Superset reasoning requires the freedom to act, and European options do not have it.

11.2 UPPER AND LOWER BOUNDS

Before any question of what an option is worth, there is the question of what it cannot be worth.

Notation. Throughout this chapter c and p are the prices of a European call and put, and C and P the prices of the American call and put on the same stock with the same strike K and expiration T; S_0 is the current stock price and r the risk-free rate. A capital P standing alone is always the American put; the bond price of later chapters carries its two arguments, $P(t, T)$.

The assumption behind everything that follows. The chapter assumes no arbitrage, as defined at the opening of Chapter 5, and frictionless trading (borrowing and lending at r, shorting without cost, no bid–ask spread), which Section 11.12 relaxes. Each bound below is proved the same way: suppose it fails, write down the trade that then costs nothing and cannot lose, and conclude the prices cannot stand.

Upper bounds. A call gives the right to buy the stock, so it can never be worth more than the stock: $c \le S_0$ and $C \le S_0$. If it were, selling the call and buying the stock would collect $c - S_0 > 0$ today, and the share covers any exercise. A put gives the right to sell at K, so it can never be worth more than K; and since a European put pays at most K at time T, its value is capped by the present value: $p \le Ke^{-rT}$, while $P \le K$. An American option can be exercised at once, so it is never worth less than its intrinsic value: $C \ge \max(S_0 - K, 0)$ and $P \ge \max(K - S_0, 0)$.

Lower bound for a European call. Four portfolios, defined once here, carry every proof in the chapter:

Portfolio A	one European call, plus cash of Ke^{-rT}
Portfolio B	one share
Portfolio C	one European put, plus one share
Portfolio D	cash of Ke^{-rT}

Compare A with B. The cash in A grows to exactly K by time T. If $S_T > K$ the call is exercised, using the K to buy the share, and A is worth S_T. If $S_T \le K$ the call lapses and A is worth K. So A is worth $\max(S_T, K)$, which is never less than B's S_T. A portfolio that is never worth less at T must not cost less today, so $c + Ke^{-rT} \ge S_0$, and since $c \ge 0$,

$$c \ge \max\left(S_0 - Ke^{-rT},\ 0\right) \tag{11.1}$$

Comparing Portfolio C, worth $\max(S_T, K)$, with Portfolio D, worth K, gives

$$p \ge \max\left(Ke^{-rT} - S_0,\ 0\right) \tag{11.2}$$

If dividends with present value D are paid during the option's life, the share collects them and the option does not, so S_0 becomes $S_0 - D$ in both bounds (Table 11.2).

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Figure 11.2 draws the region a call's bounds permit without dividends, and Example 11.1 trades a call priced below it.

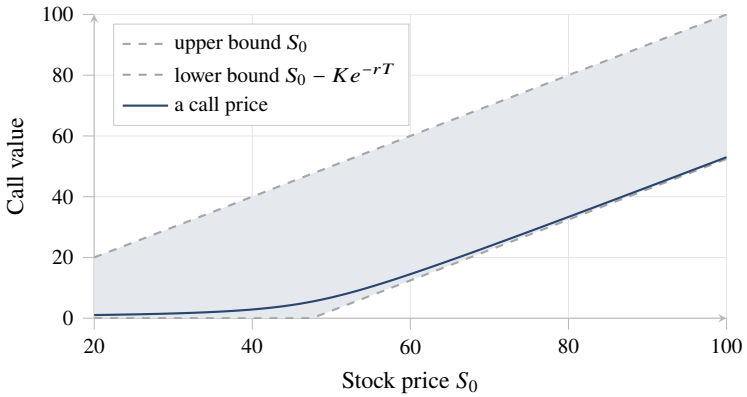


Figure 11.2: Permitted region. $K = 50$, $r = 5\%$, $T = 1$.

Example 11.1 Arbitrage when the lower bound fails

A non-dividend-paying stock trades at \$20. A one-year European call struck at \$18 trades at \$3.00. The rate is 10%.

The lower bound (11.1) is

$$20 - 18e^{-0.10} = 20 - 18 \times 0.904837 = 20 - 16.287 = \$3.713$$

so \$3.00 is too cheap. Buy the call and short the stock:

$$\text{cash received} = 20 - 3 = \$17, \qquad \text{grown for one year} = 17e^{0.10} = \$18.788$$

At expiration there are two cases. If $S_T > 18$, exercise the call, pay \$18 for the share, return it, and keep $18.788 - 18 = \$0.788$. If $S_T \le 18$, let the call lapse, buy the share in the market for $S_T \le 18$, return it, and keep at least \$0.788.

The profit is at least \$0.788 in every state, from an initial investment of nothing. The mispricing is $3.713 - 3.00 = \$0.713$, and $0.713 \times e^{0.10} = \$0.788$. The arbitrage profit is exactly the mispricing carried forward at the risk-free rate, a useful check on any such calculation.

11.3 BOUNDS ACROSS STRIKES

The bounds so far constrain one option. Three further relations constrain options at *different* strikes on the same underlying and maturity, and they matter disproportionately: they are the conditions a whole option chain must satisfy, and Chapter 19 will use them as the test of whether a volatility surface is arbitrage-free.

Write $c(K)$ for the price of a call struck at K, with $K_1 < K_2 < K_3$.

Monotonicity. $c(K_1) \ge c(K_2)$. A call with a lower strike confers the right to buy more cheaply, so it cannot be worth less. If it were, buy the low-strike call, sell the high-strike one, collect the difference, and hold a spread whose payoff is never negative.

The spread bound. $c(K_1) - c(K_2) \le (K_2 - K_1)e^{-rT}$. The payoff difference between the two calls never exceeds $K_2 - K_1$, so their price difference cannot exceed its present value.

Convexity. For equally spaced strikes, $K_2 = (K_1 + K_3)/2$,

$$c(K_1) + c(K_3) \ge 2\,c(K_2) \tag{11.3}$$

Buying the outer two calls and selling two of the middle one is a **butterfly**, and its payoff at expiration is zero outside $[K_1, K_3]$ and positive inside, peaking at K_2. A position whose

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payoff is never negative cannot be established for a credit, which is exactly what (11.3) says. With monotonicity, this makes $c(K)$ a convex, decreasing function of K, the shape drawn in Figure 11.3. Both results hold for American calls too, since the long calls can be exercised whenever a short one is assigned; the spread bound loses its discount, $C(K_1) - C(K_2) \le K_2 - K_1$, because assignment can demand $K_2 - K_1$ at once.

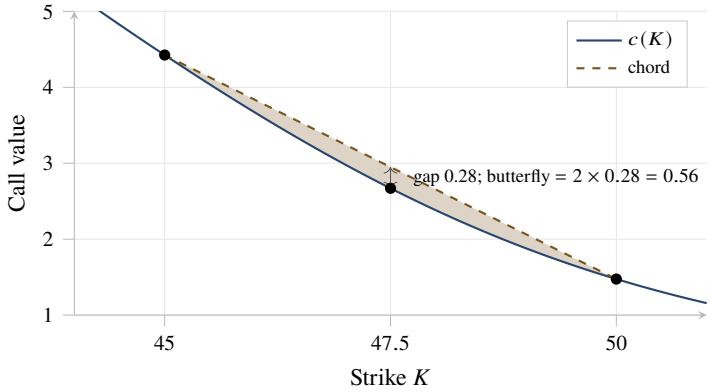


Figure 11.3: Convexity in the strike, at the three strikes of Example 11.2. Shaded: the gap between chord and curve.

Geometrically, convexity says the curve lies below any chord joining two of its points. The vertical gap between the chord and the curve at K_2 is half the value of the butterfly, since the butterfly sells the middle call twice, and it must be nonnegative.

Example 11.2 Testing a chain for convexity

Take the calls from Table 10.2, using mid prices. Options on individual US stocks are American, and monotonicity and convexity hold for both styles, so write $C(45) = 4.425$, $C(47.50) = 2.67$, $C(50) = 1.475$. The strikes are equally spaced, so (11.3) requires

$$4.425 + 1.475 = 5.90 \;\ge\; 2 \times 2.67 = 5.34$$

The chain passes, with \$0.56 to spare, which is the butterfly's value.

Check the other two conditions as well. Monotonicity holds: 4.425 > 2.67 > 1.475. Because the options are American, the spread bound takes its undiscounted form:

$$C(45) - C(50) = 2.95 \;\le\; 50 - 45 = \$5$$

with about \$2 to spare.

Now suppose the 47.50 call were instead quoted 3.00 bid / 3.10 ask. Convexity would fail: $2 \times 3.05 = 6.10$ against 5.90. Trade it at the quotes: buy the 45 call at its \$4.50 ask, buy the 50 call at its \$1.51 ask, sell two 47.50 calls at the \$3.00 bid:

$$\text{net cash} = 2 \times 3.00 - 4.50 - 1.51 = 6.00 - 6.01 = -\$0.01$$

That is a penny of cost, not a credit, so the violation at mid prices does *not* survive the bid–ask spread, and there is no trade. Had the 47.50 bid been \$3.05, the position would have been established for a \$0.09 credit with a guaranteed payoff between \$0 and \$2.50: an arbitrage.

11.4 PUT–CALL PARITY

The chapter's central result relates the call and the put at the same strike and maturity, and it compares Portfolios A and C of Section 11.2. At expiration, A is worth $\max(S_T, K)$ as shown there. Portfolio C is worth S_T if $S_T > K$ (the put lapses) and K if $S_T \le K$ (the put is exercised),

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also $\max(S_T, K)$. Two portfolios worth the same at T in every state must cost the same today:

$$\boxed{c + Ke^{-rT} = p + S_0} \tag{11.4}$$

This is **put–call parity**. It holds for European options on a non-dividend-paying stock, and it requires no assumption whatever about how the stock behaves. If dividends with present value D are expected during the option's life, the share in Portfolio C also delivers those, and the relation becomes

$$c + D + Ke^{-rT} = p + S_0 \tag{11.5}$$

Because $c - p$ is linear in K in either form, a put butterfly is worth what the call butterfly at the same strikes is worth, so puts are convex in the strike too, and the call's monotonicity and spread bound become $0 \le p(K_2) - p(K_1) \le (K_2 - K_1)e^{-rT}$. Example 11.3 shows what a violation of (11.4) is worth and how it is traded.

Example 11.3 Trading a parity violation

A stock trades at \$31 and pays no dividends. Three-month European options struck at \$30 are quoted: the call at \$3.00, the put at \$2.25. The rate is 10%.

Parity (11.4) says the put should be

$$p = c + Ke^{-rT} - S_0 = 3.00 + 30e^{-0.025} - 31 = \$1.2593$$

The put is quoted \$0.99 too high. Sell it, and hedge by buying the replicating portfolio: buy the call and short the stock, investing the proceeds.

$$\text{cash in} = 2.25 - 3.00 + 31 = \$30.25, \qquad 30.25\, e^{0.025} = \$31.0158$$

At expiration, if $S_T > 30$, exercise the call. If $S_T \le 30$, the put is exercised against you and you buy at \$30. Either way the short is closed for \$30, leaving

$$31.0158 - 30 = \$1.0158$$

in every state. Again this is the mispricing carried forward: $0.9907 \times e^{0.025} = \$1.0158$.

This trade has a name. Long call, short put, short stock is a **reversal**; the opposite is a **conversion**.

Parity for other underlyings

Chapter 10 cataloged options on indices, currencies, and futures. Parity extends to each of them by replacing the share in Portfolio C with whatever position replicates the underlying at time T, and the results are worth having explicitly:

$$\text{Index, dividend yield } q: \quad c + Ke^{-rT} = p + S_0 e^{-qT} \tag{11.6}$$

$$\text{Currency, foreign rate } r_f: \quad c + Ke^{-rT} = p + S_0 e^{-r_f T} \tag{11.7}$$

$$\text{Futures, price } F_0: \quad c + Ke^{-rT} = p + F_0 e^{-rT} \tag{11.8}$$

Equations (11.6) and (11.7) are the same statement: an index yielding q and a currency yielding r_f both require holding less than one unit today to have one unit at T, which is the reasoning of Sections 5.4 and 5.7 carried into option pricing.

The futures version (11.8) rearranges to something that holds for every underlying once F_0 is read as its forward price to T, and is used constantly on trading desks:

$$c - p = (F_0 - K)e^{-rT} \tag{11.9}$$

Set $K = F_0$ and the right-hand side vanishes: a call and a put struck at the forward price have equal value. This is why the strike equal to the forward price, called **at-the-money forward**, is the usual reference point for quoting options, and why a straddle at that strike (Chapter 12) is

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the standard way to trade volatility without taking a directional position: the call and the put cost the same, and their deltas (Chapter 18) nearly offset. Chapter 19 measures moneyness and log-moneyness from this strike.

11.5 SYNTHETIC POSITIONS

Equation (11.4) has four terms, so it can be rearranged four ways, and each rearrangement is a recipe for manufacturing one instrument from the others:

$$S_0 = c - p + Ke^{-rT} \qquad \text{synthetic long stock} \tag{11.10}$$

$$c = p + S_0 - Ke^{-rT} \qquad \text{synthetic call}$$

$$p = c - S_0 + Ke^{-rT} \qquad \text{synthetic put}$$

$$-S_0 = p - c - Ke^{-rT} \qquad \text{synthetic short stock} \tag{11.11}$$

Figure 11.4 shows the most useful of these graphically: a long call combined with a short put at the same strike has exactly the payoff of a long forward.

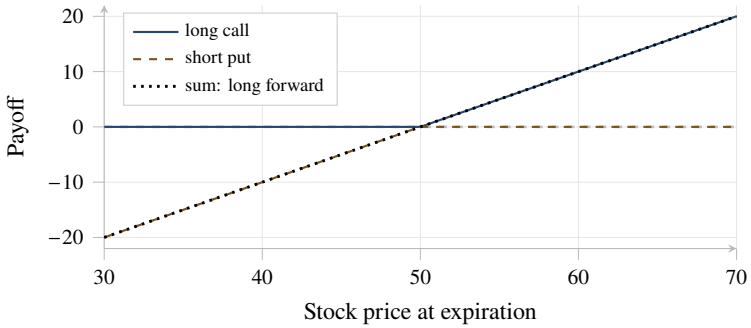


Figure 11.4: Long call plus short put at the same strike. $K = 50$.

Equation (11.11) is the synthetic short Section 10.7 promised, whose put trades above parity by the borrow fee (Example 10.4).

11.6 READING THE IMPLIED FORWARD

Parity is usually taught as a test. In practice it is a *measurement device*: rearranged, it extracts from observable option prices a quantity no one quotes directly and everyone needs.

Rearranging (11.9) gives the option market's own forward price,

$$F_0 = K + (c - p)\, e^{rT} \tag{11.12}$$

It requires no view on dividends, no borrow-cost assumption, and no model.

Example 11.4 What the chain says about dividends

Take the 47.50 strike from Table 10.2: call mid 2.67, put mid 2.575, 45 days to expiration, $r = 5\%$. From (11.12),

$$F_0 = 47.50 + (2.67 - 2.575)\, e^{0.05\times 45/365} = 47.50 + 0.095 \times 1.006183 = \$47.596$$

The spot price is \$48.30, so the option market's 45-day forward is 70 cents below spot. That is striking, because interest alone would put the forward *above* spot, by $48.30(e^{0.0061644} - 1) = \$0.30$. Something is pulling it down by a dollar.

That something is mostly dividends, the roughly \$1.00 that Example 10.3 found from the same quotes (1.092 − 0.095). So the chain says the market expects about \$1.00 of dividends before expiration, roughly 2% of the share price in 45 days, which suggests a high-yielding stock with an

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ex-dividend date inside the option's life. The option prices contain the market's forecast, and where that forecast differs from the announced dividend, the gap is itself information: about a dividend the market expects to change, or, as the second caveat explains, about the cost of borrowing the stock.

Two caveats keep this honest. First, the options of Table 10.2 are American, and Section 11.11 shows that arbitrage alone confines $C - P$ here only to a band about \$1.29 wide. The reading is usable because at 47.50, nearest the money, both options are liquid and neither carries much early-exercise premium, so $C - P$ stays close to the European $c - p$; the \$1.00 is good to ten cents or so, not to the cent. Second, parity gives *one* equation, and the forward embeds dividends, borrow costs, and financing together. Having computed \$1.00, one cannot separate "dividends of \$1.00" from "dividends of \$0.60 plus a 40-cent borrow cost" without outside information. Practitioners therefore quote the implied forward and stop there: decomposing it further is a modeling choice, not a measurement.

11.7 BOX SPREADS: PARITY AS A FINANCING TRADE

Combining a synthetic long at one strike with a synthetic short at another produces something with no equity exposure at all. Buy the call and sell the put at K_1; sell the call and buy the put at K_2, with $K_1 < K_2$. From (11.10) and (11.11) the stock cancels, and the payoff at expiration is $K_2 - K_1$ regardless of where the stock finishes. This is a **box spread**, and it is a zero-coupon bond assembled from four options; Example 11.5 reads off the interest rate its price implies.

Example 11.5 Borrowing and lending through the option market

An investor constructs a box on a stock index using one-year European options at strikes 4,900 and 5,000. Whatever the index does, the position pays 100 index points at expiration, \$10,000 per contract at the \$100 multiplier.

If the box can be bought for 95.12 index points (\$9,512 per contract), the implied lending rate is

$$r = \ln\left(\frac{100}{95.12}\right) = \ln(1.051304) = 5.00\%$$

If it can be bought for 93.00 points, the rate is $\ln(100/93.00) = 7.26\%$, and the investor lends at 7.26% on a fully collateralized, centrally cleared position. *Selling* the box borrows at that rate.

Institutions use box spreads as a financing instrument, competing with repo, because the position is cleared, requires no counterparty credit assessment, and can be executed in large amounts on an exchange. The implied rate is a directly observable financing rate, worth comparing against the rates of Chapter 4, since discrepancies between them are the option market's version of the basis trades of Chapter 6.

Case Study 11.1 The box that was not riskless

A box spread is riskless *if the options are European*. In 2019 a retail trader on a US brokerage constructed box spreads using options on an ETF (which, as Section 10.3 noted, are American) and treated the position as a guaranteed loan. The account was reported to have lost more than ten times its \$5,000 of capital.

The mechanism is exactly the assignment risk of Section 10.9. Two of the four legs are short options. If the ETF moves so that one of the short options is deep in the money, its holder may exercise it early. The trader is then assigned, must deliver or receive shares immediately, and is left holding a position that no longer has the offsetting structure the box depended on—at a moment chosen by someone else, not by them.

Two lessons follow, and both generalize past this episode. First, every result so far must be re-examined before it is applied to American options. Second, a position that is

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riskless at expiration is not necessarily riskless *before* expiration. Chapter 3 made this point about funding and Chapter 6 about the cash–futures basis trade. Here it appears as a counterparty's right to exercise early, when it suits them rather than you.

11.8 EARLY EXERCISE: AMERICAN CALLS WITHOUT DIVIDENDS

An American call on a non-dividend-paying stock should never be exercised early. The result is stronger than it first appears, and it has two independent proofs worth knowing because they generalize differently.

The bounds argument. From (11.1), an American call is worth at least $S_0 - Ke^{-rT}$, since it is worth at least as much as the European call. Exercising yields only $S_0 - K$. Because $Ke^{-rT} < K$ whenever $r > 0$,

$$\underbrace{S_0 - Ke^{-rT}}_{\text{value if sold}} > \underbrace{S_0 - K}_{\text{value if exercised}}$$

The holder always does better selling the option than exercising it. Note what this proves: at every date before expiration the call is worth more alive than exercised, so the right to exercise early is never used and adds nothing, and $C = c$ for a non-dividend-paying stock.

The economic argument. Exercising early does two things, both bad. It pays K now rather than at T, giving up the interest, $K - Ke^{-rT} = K\left(1 - e^{-rT}\right)$, which is the difference between the two sides of the display above. And it discards insurance: a holder who exercises owns the stock outright and suffers fully if it falls, whereas a holder who waits keeps a floor at zero. The second point is why the result holds even at $r = 0$, where the interest argument vanishes. Example 11.6 puts numbers on both.

Example 11.6 Why not to exercise

A stock trades at \$50 and pays no dividend. An investor holds an American call struck at \$40 with one month to run. The rate is 6%. Exercising immediately yields $50 - 40 = \$10$.

But the call must be worth at least

$$50 - 40e^{-0.06\times 1/12} = 50 - 40 \times 0.995012 = 50 - 39.800 = \$10.20$$

so selling it yields at least \$10.20: twenty cents more, guaranteed, with no view on the stock.

Suppose instead the investor wants to own the stock and holds the call for that reason. Even then, exercising is wrong: they should sell the call for at least \$10.20 and buy the stock at \$50, which costs \$39.80 net against \$40 for exercising. And if what they want is to keep the exposure for the remaining month rather than own the stock, holding the call is better still, because exercising would also throw away the protection against the stock falling below \$40 during that month.

In a frictionless market no preference makes exercising right, which is unusual: most results in finance depend on what the investor wants.

11.9 EARLY EXERCISE: AMERICAN PUTS

For puts the conclusion reverses, and seeing why shows what the call case relied on.

Take the extreme case. A stock has fallen to \$0.05 and an American put struck at \$50 has six months to run. Exercising now yields \$49.95 immediately. Since the stock cannot fall below zero, waiting can add at most the last five cents, and it costs interest on the strike for as long as the holder waits: held to expiration, the \$50 is worth $50e^{-rT}$ today, less than \$49.95 at

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any rate above about 0.2%. Waiting a shorter time pays only if the stock reaches zero almost at once, which no realistic model allows. Exercise now.

The general principle is that exercising a put brings forward the receipt of K, which earns interest, whereas exercising a call brings forward the payment of K, which costs interest. The insurance argument still cuts the other way (exercising discards the chance of the stock falling further), but for a sufficiently deep in-the-money put the interest gain dominates.

There is therefore a **critical stock price** $S^*(t)$ below which immediate exercise is optimal, and it rises toward K as expiration approaches. Figure 11.5 shows the consequence for value.

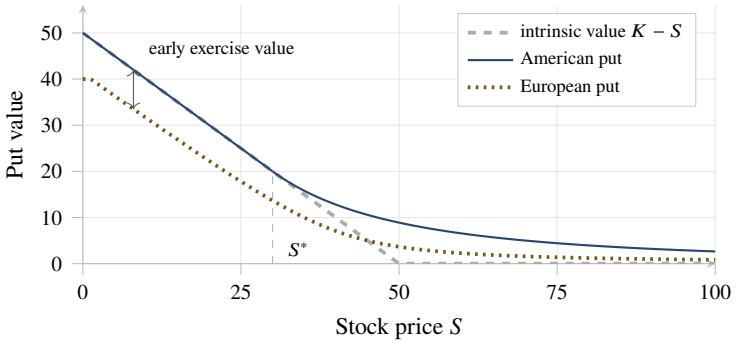


Figure 11.5: Put values against the stock price. $K = 50$; schematic, with the gap between the two exaggerated.

Three features of Figure 11.5 carry the argument. The American put coincides with intrinsic value below S^*, where it is exercised. The European put falls *below* intrinsic value for low stock prices (it is worth less than $K - S$ because the K arrives only at T), which is possible precisely because it cannot be exercised. And the European put is capped at Ke^{-rT} while the American put is capped at K. The gap between the two curves is the value of the right to exercise early, and unlike the call case it is genuinely positive: $P > p$.

11.10 DIVIDENDS AND EARLY EXERCISE

Now consider the third question Chapter 10 deferred. Ordinary cash dividends do not adjust the contract (Section 10.2), so an option holder simply suffers the price drop on the ex-dividend date. In a frictionless market that, and only that, can make early exercise of an American call rational. Frictions supply a second reason: a stock that is costly to borrow earns its owner a lending fee that a call holder does not receive, so it behaves like a stock paying a dividend, and a deep in-the-money call on it can be worth exercising early.

The trade-off is sharp. Exercising just before the stock goes ex-dividend captures the dividend D, because the exerciser owns the stock across the ex-date. The costs are the interest forgone on K over the remaining life $T - t$, where t is just before the ex-dividend date (only up to the next ex-dividend date, if one falls before expiration), and the downside protection given up. Comparing the dividend with the interest alone gives a clean condition: early exercise cannot be optimal unless

$$D > K\left(1 - e^{-r(T-t)}\right) \tag{11.13}$$

Example 11.7 applies the condition to a holder facing an ex-dividend date.

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Example 11.7 Should the call be exercised for the dividend?

A stock trades at \$60 and goes ex-dividend tomorrow, paying \$1.50. An investor holds American calls struck at \$55 with two months remaining. The rate is 5%.

The interest forgone by paying the strike two months early is

$$K\left(1-e^{-r(T-t)}\right) = 55\left(1-e^{-0.05\times 2/12}\right) = 55\,(1-0.991701)$$
$$= 55\times 0.008299 = \$0.456$$

The dividend is \$1.50, comfortably larger. Condition (11.13) is satisfied and exercise may well be optimal: capturing \$1.50 outweighs giving up \$0.46 of interest.

Now change one number. If the dividend were \$0.30 instead, the condition fails (\$0.30 against \$0.456) and exercise is *definitely* wrong, whatever else is true.

Condition (11.13) is necessary, not sufficient. When it holds, exercise *may* be optimal, but the comparison above ignored the insurance the holder abandons, which is worth something whenever the stock might fall below \$55 in the remaining two months. A holder deep in the money, where that protection is nearly worthless, should exercise; a holder near the money should think harder.

The practical rule is that early exercise of American calls, when it happens at all, happens immediately before an ex-dividend date and essentially never otherwise, which is why option desks maintain dividend calendars.

For American puts, dividends work the opposite way. A dividend lowers the stock price, which raises the put's value and makes waiting more attractive. Dividends therefore *discourage* early exercise of puts, and the critical price S^* falls as a dividend approaches.

11.11 PARITY BOUNDS FOR AMERICAN OPTIONS

Put–call parity is an equality for European options and fails for American ones, because either side may be exercised at a time the arbitrage argument does not control. What survives is a pair of inequalities. For a non-dividend-paying stock,

$$S_0 - K \;\le\; C - P \;\le\; S_0 - Ke^{-rT} \tag{11.14}$$

For the right-hand inequality, hold Portfolio C and be short Portfolio A, both in American form: hold the put and one share, write the call, and borrow Ke^{-rT}. If the call is assigned early, the share is delivered for K, which more than repays the loan, and the put is kept; otherwise everything nets to zero at T, and any dividends collected along the way are extra. So $P + S_0 - C - Ke^{-rT} \ge 0$, with or without dividends. The left-hand inequality follows from a portfolio argument of the same kind. With dividends of present value D, only the left edge moves, down by D:

$$S_0 - D - K \;\le\; C - P \;\le\; S_0 - Ke^{-rT} \tag{11.15}$$

The practical content is that $C - P$ sits in a band of width $K - Ke^{-rT}$, the interest on the strike over the option's life, plus D when the stock pays a dividend. Without dividends a short-dated band is narrow and parity is very nearly an equality. For the Table 10.2 stock at the 47.50 strike, with D = \$1.00, the band runs from $48.30 - 1.00 - 47.50 = -\$0.20$ to $48.30 - 47.208 = \$1.09$, \$1.29 wide; the European value $S_0 - D - Ke^{-rT}$ is \$0.092, and the quoted $C - P$ of \$0.095 sits almost exactly on it. Table 11.2 collects every bound in the chapter: the European difference is a point, and the American one a band around it.

11.12 WHAT VIOLATIONS ACTUALLY MEAN

Everything above assumed frictionless trading. Real chains appear to violate these relations routinely, and reading those violations correctly is the practical skill this chapter is for.

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Table 11.2: Bounds from arbitrage alone; D is the present value of dividends (0 if none).

	Lower bound	**Upper bound**
European call c	$\max(S_0 - D - Ke^{-rT},\ 0)$	S_0
American call C	$\max(S_0 - D - Ke^{-rT},\ S_0 - K,\ 0)$	S_0
European put p	$\max(D + Ke^{-rT} - S_0,\ 0)$	Ke^{-rT}
American put P	$\max(D + Ke^{-rT} - S_0,\ K - S_0,\ 0)$	K
$c - p$	$S_0 - D - Ke^{-rT}$	$S_0 - D - Ke^{-rT}$
$C - P$	$S_0 - D - K$	$S_0 - Ke^{-rT}$

Borrow costs. A borrow fee opens a band above parity (Example 10.4). Inside it there is no arbitrage, and puts priced above parity are quoting the borrow fee, not a bearish view.

Transaction costs. A conversion involves four legs (call, put, stock, financing), each crossing a spread. Table 10.2's spreads at the 50 strike, \$0.07 on the call and \$0.15 on the put, cost about \$0.11 against mid prices on the option legs alone, and twice that if the position is unwound early: more than many apparent violations.

The financing rate is not r. Every relation in this chapter used the risk-free rate. Section 5.15 argued at length that no participant borrows at it, and the relevant rate is the one *this* participant can actually finance at. Two firms observing the same chain can correctly disagree about whether an arbitrage exists.

Dividends are estimates. Parity with dividends, (11.5), requires knowing D. For an option spanning an uncertain dividend, the apparent violation reflects a different dividend forecast rather than a mispricing.

The lesson is the one that recurs from Chapter 5: *these relations define a band, not a point*, and the width of the band is a real economic quantity. A trader who assumes it is zero and trades every apparent violation will discover the width empirically, which is more expensive.

11.13 WHO TRADES THESE RELATIONSHIPS

Several kinds of participant trade these relationships directly, and who they are explains why real chains are as consistent as they are.

Option market makers are the largest users. A market maker who has accumulated an unwanted position uses conversions and reversals (Section 11.4) to transform it (turning a short put into a long stock position plus a short call) at a cost they can compute in advance. This is not speculation; it is inventory management, and it is why quotes across a chain move together rather than independently.

Dispersion and volatility desks (Chapter 22) use parity to strip the directional component out of a volatility position.

Financing desks trade box spreads (Section 11.7); when box rates diverge from repo rates, the reason is usually balance-sheet capacity rather than mispricing, as Section 5.15 argued.

Structured-product desks use synthetics (11.10)–(11.11) to manufacture exposures they cannot trade directly. Where shorting is restricted, prohibited, or prohibitively expensive, the synthetic short is the only route.

Index and dividend traders use the implied forward of Section 11.6 as a traded quantity in its own right. Dividend expectations extracted from option chains are traded directly in dividend futures and dividend swaps, which pay the dividends actually distributed over a stated period; there the number computed in Example 11.4 is the price being traded.

The common thread is that none of these participants is forecasting the stock. They are trading the relationships among instruments written on it, and the fact that they compete to do so is the mechanism that makes the relationships hold.

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SUMMARY

Every result in this chapter follows from the absence of arbitrage (a strategy that costs nothing, cannot lose, and can gain), without any model of how the stock behaves. The book assumes throughout that no such opportunity exists, as a statement about prices rather than about people.

Six factors determine an option's value. Calls and puts both rise with volatility, because the payoff is truncated at zero and greater dispersion carries no symmetric penalty; higher rates raise calls and lower puts; and more time helps American options unambiguously but not necessarily European ones, which cannot act on the extra time. Arbitrage alone then bounds each price above and below, and across strikes it imposes monotonicity, the spread bound and convexity. Convexity says a butterfly cannot have negative value, a test that must be run at prices one can trade rather than at mid prices, and Chapter 19 uses it to test a volatility surface for arbitrage.

Put–call parity, $c + Ke^{-rT} = p + S_0$, links the call and put at the same strike and maturity. Rearranged, it manufactures synthetic stock and synthetic options, and the synthetic short explains why puts trade above parity on hard-to-borrow stocks; two synthetics at different strikes make a box spread, a zero-coupon bond built from options, riskless only when the options are European, and used as a financing instrument. Parity is also a measurement: the implied forward read off an option chain is the chain's most useful single number, and its shortfall from the carried-forward spot reflects mostly expected dividends, mixed with borrow costs.

An American call on a non-dividend-paying stock should never be exercised early, because selling always beats exercising: the option is worth more than its exercise value, and exercising discards downside protection. An American put may be exercised early, because deep in the money the interest earned on receiving the strike now outweighs the option value given up, so below a critical stock price exercise is optimal. In a frictionless market, dividends are the only reason to exercise an American call early, and when they make it optimal, it happens immediately before an ex-dividend date.

For American options parity becomes a pair of inequalities rather than an identity. In practice all these relations define bands rather than points, widened by borrow costs, transaction costs, financing that is not at the risk-free rate, and uncertain dividends. The width is an economic quantity worth measuring, not an error term to ignore.

FURTHER READING

Merton's 1973 paper on rational option pricing established most of the results in this chapter and states them in full. Stoll's 1969 paper introduced put–call parity to the modern literature. Natenberg's *Option Volatility and Pricing* describes conversions, reversals, and boxes as a market maker trades them. Ofek, Richardson and Whitelaw (2004) is the evidence for Section 11.12: apparent parity violations track borrow costs and short-sale constraints. Jensen and Pedersen (2016) is the place to start on when frictions make early exercise of a call rational, and van Binsbergen, Diamond and Grotteria (2022) read risk-free rates off index-option prices through the parity relation a box spread locks in.

PRACTICE QUESTIONS

11.1 Explain why both a call and a put increase in value with volatility, and why a forward contract's value does not depend on volatility at all.

11.2 A non-dividend-paying stock trades at \$45. A six-month European call struck at \$42 trades at \$3.10 and the rate is 6%. Is the lower bound violated? If so, construct the arbitrage and compute the profit.

11.3 A stock trades at \$28 with no dividends. Four-month European options struck at \$27 are

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quoted: call \$2.60, put \$1.10, rate 5%. Test parity and, if it fails, state the trade and the profit.

11.4 Using put–call parity, express a synthetic long put in terms of a call, the stock, and cash. Verify the payoff in both states at expiration.

11.5 Explain why an American call on a non-dividend-paying stock should never be exercised early, giving both the bounds argument and the economic argument. Which of the two still works when $r = 0$?

11.6 A stock has fallen to \$0.25 and an American put struck at \$40 has three months remaining. Should it be exercised? Give the numerical comparison with $r = 4\%$.

11.7 A stock at \$80 goes ex-dividend tomorrow, paying \$0.90. An investor holds American calls struck at \$75 with three months remaining; the rate is 4%. Apply condition (11.13) and state what conclusion is and is not justified.

11.8 Construct a box spread from options struck at \$40 and \$50 expiring in six months, list the four legs, and state the payoff. If the box costs \$9.70, what rate is implied?

11.9 State the three relations that constrain call prices at different strikes on the same underlying and maturity, and for each identify the position that profits if it is violated.

11.10 Using the mid prices of Table 10.2 at every strike, with $r = 5\%$, compute the implied forward (11.12) and the dividend it implies, $S_0 - F_0 e^{-rT}$. At which strikes does the reading drift, in which direction, and why? Which strike would you quote?

11.11 State the parity bounds for American options, explain why the European equality becomes a band, identify which side of the band binds when the put is deep in the money and why, and state how the band changes when the stock pays dividends of present value D.

11.12 A chain shows puts trading 40 cents above parity on a \$50 stock with six months to expiration. Using Example 10.4, state the borrow fee that would account for it, and explain why observing a violation of this size is not the same as being able to profit from it.

FURTHER QUESTIONS

11.13 Section 11.1 states that more time to expiration can *reduce* the value of a European option. Construct two explicit examples, one for a call and one for a put, in which this occurs, and identify precisely which feature of European exercise makes it possible.

11.14 The case study in Section 11.7 describes a box spread that was not riskless because it used American options. Work out exactly which of the four legs can be assigned early, under what stock price conditions, and what position the trader is left holding. Then state a rule for when a box may safely be treated as a financing trade.

11.15 Condition (11.13) is necessary but not sufficient. Construct a case in which the condition holds and early exercise is nevertheless wrong, and explain what quantity the condition omits. How would you decide the question in practice?

11.16 An option chain on a hard-to-borrow stock shows puts persistently above parity by an amount implying a 15% borrow fee. A colleague proposes selling puts and buying calls to "harvest the mispricing," delta-hedging with stock in the manner of Chapter 18. Explain what exposure this position actually carries, and what would have to happen for it to lose money.

11.17 Derive the left-hand inequality in (11.14) by constructing two portfolios and comparing their values. State which step uses the put's right to early exercise, and show that for European options the inequality follows from the equality (11.4).

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Chapter 12
Trading Strategies Involving Options

Options can be combined into an unlimited number of payoff shapes, and textbooks that catalog them tend to leave readers able to draw a great many diagrams and unable to say why anyone would want one. This chapter is organized around the opposite question. Every position is a bet on something, and a well-chosen strategy is one whose bet matches the view its holder actually has.

There are only four things to bet on. **Direction**—where the underlying goes. **Magnitude**—how far it moves, regardless of which way. **Time**—whether the move happens before the option expires. And **skew**, the relative price of downside and upside protection. Most strategies are a bet on two or three of these simultaneously, which is why traders lose money on positions whose directional view turns out to be right.

Every one of these bets is levered. A spread that costs $2.04 on a $48.30 stock earns up to 1.45 times its cost if the stock rises 8.7%, and loses all of it if the stock falls 1.7%. Leverage is not a fifth thing to bet on; it is the multiplier on the other four, the same multiplier Example 1.5 measured for futures, and it is why a position whose diagram looks tame can be anything but tame in a portfolio. The examples below therefore report what each position costs alongside what it can pay, because the ratio of the two is where the leverage sits.

Chapter 11 has already supplied the tools. Put–call parity means many strategies that appear distinct are the same position in a different form, and Section 12.1 begins there because it eliminates a great deal of apparent complexity before any of it accumulates.

Sections 12.2 to 12.6 take the strategies by the view they express, from spreads for direction to risk reversals for skew; the rest of the chapter prices one view four ways, builds arbitrary payoffs, turns to the institutional uses (the collar and the portfolio hedge), and closes with structured products, position management and costs.

In the worked examples, the numbers come from the option chain of Table 10.2 (a stock at $48.30, 45 days to expiration), and they are computed at the *bid and ask*, not the mid. Section 12.14 shows why that decision changes the conclusions.

12.1 STRATEGIES THAT ARE EQUIVALENT

Two of the most widely used retail strategies are not new positions at all.

A **covered call** is long stock plus a written call. Rearranging parity (11.4), which holds exactly for European options,

$$S_0 - c = Ke^{-rT} - p$$

so the covered call has the payoff of a *written put* plus cash. A **protective put** is long stock plus a purchased put, and the same rearrangement gives

$$S_0 + p = c + Ke^{-rT}$$

which is a *purchased call* plus cash. Figure 12.1 shows both.

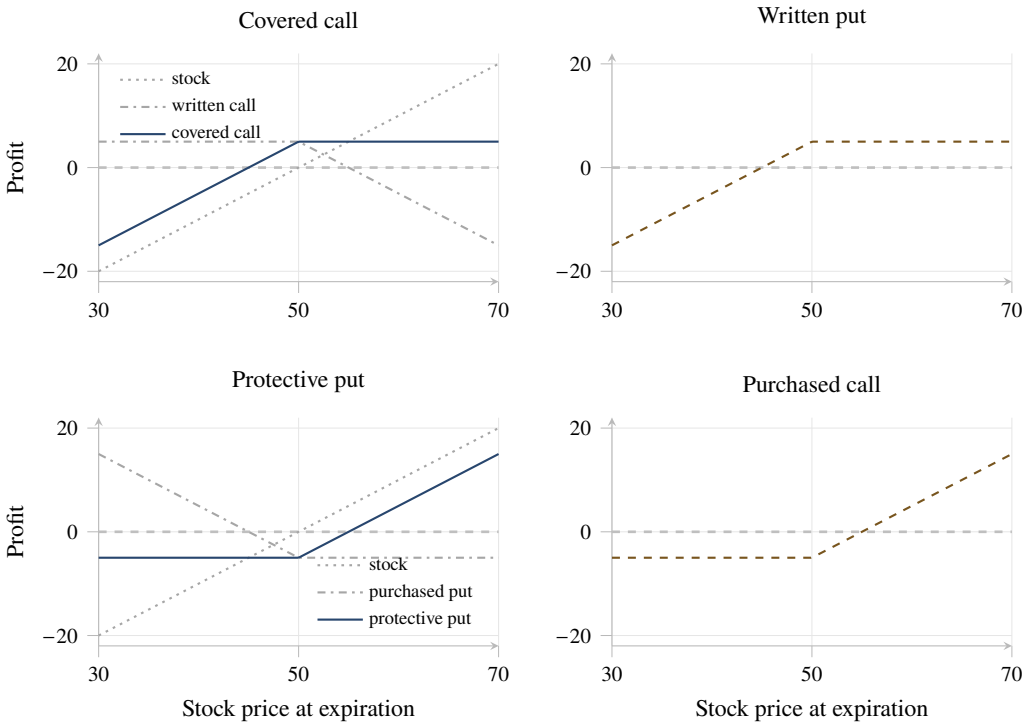


Figure 12.1: Profit at expiration of each strategy, with its components in gray, and of its parity equivalent. $K = 50$.

The panels in each row are identical, and this is not a curiosity: it has three practical consequences that recur through the chapter.

It disciplines the marketing. A covered call is often sold as a conservative income strategy and a written put as a speculative one. Held to expiration, the covered call is a written put plus cash, a **cash-secured put**, and brokers, who set options-approval levels, commonly approve the two at the same lowest level; the higher level is for the *naked* put, the same exposure without the cash. The *market risk* is identical.

It tells you which to trade. If two routes to the same exposure exist, take the cheaper. Writing the put is one transaction rather than two, so it pays the bid–ask spread once instead of twice.

It identifies the real bet. A covered call writer is short a put: they are selling insurance, and they profit when the market is calm and lose when it falls. Describing this as "generating income from a stock you already own" is accurate about the cash flow and silent about the risk.

12.2 SPREADS: A DIRECTIONAL VIEW WITH BOTH ENDS CAPPED

A **spread** combines two or more options of the same type, each of them a **leg** of the spread. The simplest is a directional bet financed by selling away the part of the move you do not expect.

A **bull call spread** buys a call and sells a higher-strike call. It costs less than the call alone, because the sale pays for part of it, and in exchange the profit stops at the higher strike. A **bear put spread** is the mirror image (Example 12.2). Each can also be built from the other type of option: a **bull put spread** sells a put and buys a lower-strike put, and a **bear call spread** sells a call and buys a higher-strike call. These bring in a net premium, so for each view one version is a **debit** spread, which costs a net premium to establish, and the other a **credit** spread.

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By put–call parity the two versions of the same view have the same profit diagram apart from interest on the strike difference, so the choice between them is about cash up front and margin, not about the view; put credit spreads are a staple of investors who sell options for income.

Example 12.1 A bull call spread

Using Table 10.2, buy the 47.50 call at its \$2.72 ask and sell the 52.50 call at its \$0.68 bid:

$$\text{net debit} = 2.72 - 0.68 = \$2.04 \text{ per share, or } \$204 \text{ per spread}$$

The outcomes are:

Stock at expiration	Payoff	Profit
≤ 47.50	0.00	−2.04
49.54	2.04	0.00
≥ 52.50	5.00	+2.96

Maximum profit \$2.96, maximum loss \$2.04, breakeven \$49.54. The stock must rise 2.6% in 45 days merely to break even, and 8.7% to earn the maximum. The **reward-to-risk ratio** (maximum profit divided by maximum loss, a trader's shorthand that says nothing about how likely either is) is 1.45 to 1.

Compare with buying the 47.50 call outright at \$2.72. That costs 33% more and has unlimited upside. The spread does better at every price below \$53.18, rise or fall, and the outright call above it. The spread is not a cheaper way to be bullish; it is a bet on a specific range.

Example 12.2 A bear put spread

The mirror image, for a trader who expects a fall. Using Table 10.2, buy the 50 put at its \$3.95 ask and sell the 45 put at its \$1.81 bid:

$$\text{net debit} = 3.95 - 1.81 = \$2.14 \text{ per share, or } \$214 \text{ per spread}$$

The outcomes are:

Stock at expiration	Payoff	Profit
≥ 50	0.00	−2.14
47.86	2.14	0.00
≤ 45	5.00	+2.86

Maximum profit \$2.86, maximum loss \$2.14, breakeven \$47.86. The stock must fall 0.9% in 45 days to break even and 6.8% to earn the maximum, and the reward-to-risk ratio is 1.34 to 1, against 1.45 to 1 for the bull spread of Example 12.1. Part of the gap is moneyness (the bull spread buys a strike \$0.80 below spot, the bear spread one \$1.70 above) and part is the chain's dividend; Example 12.5 separates the forward from the skew.

12.3 BUTTERFLIES AND CONDORS: A VIEW THAT THE PRICE STAYS PUT

Section 11.3 introduced the butterfly as the constraint that makes the call price convex in the strike. As a trade, it profits if the underlying stays near the middle strike until expiration.

Buy one call at K_1, sell two at K_2, buy one at K_3, with equal spacing. The payoff is zero outside $[K_1, K_3]$ and peaks at $K_2 - K_1$ in the middle. A **condor** spreads the two middle strikes apart, producing a plateau instead of a peak: a wider target for a lower maximum. As traded, most condors are **iron condors**: an out-of-the-money put spread and an out-of-the-money call spread, both sold, entered for a credit and, by parity, with the call condor's profit profile. This credit form is what premium-selling funds run, and Chapter 23 finds it a large and growing share of same-day index options volume.

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Figure 12.2 collects the four profit profiles, and reading them together shows what every spread buys and sells: the strike that was sold caps the profit, and the premium it brings in is exactly what makes the structure cheaper than the option bought alone.

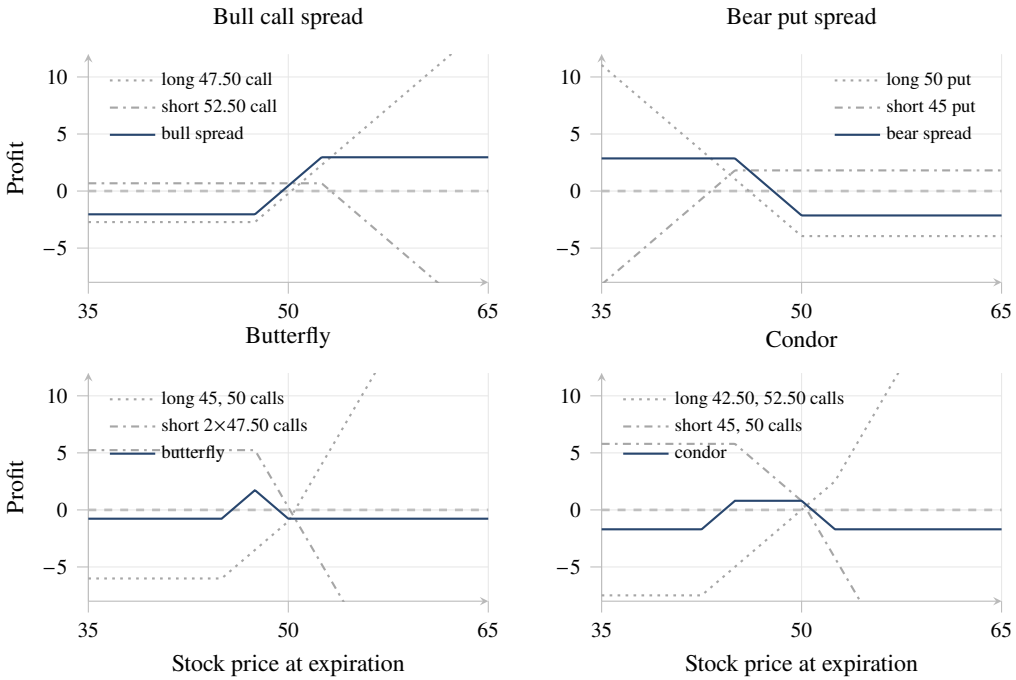


Figure 12.2: Profit at expiration for four spread structures at the quotes of Table 10.2, with the legs in gray.

Example 12.3 A butterfly, priced at the quotes

From Table 10.2, buy the 45 call at \$4.50, sell two 47.50 calls at \$2.62, buy the 50 call at \$1.51:

$$\text{cost} = 4.50 - 2 \times 2.62 + 1.51 = 6.01 - 5.24 = \$0.77$$

The maximum payoff is \$2.50, achieved if the stock finishes at exactly \$47.50, so the maximum profit is \$1.73 against a maximum loss of \$0.77, a reward-to-risk ratio of 2.25 to 1. The breakevens are \$45.77 and \$49.23.

That ratio looks attractive until the window is examined. The stock is at \$48.30, so it must finish between 5.2% below and 1.9% above its current level, in 45 days, for the position to make anything at all. Butterflies pay well precisely because that is unlikely.

Now comes the point Section 11.3 set up. At mid prices Example 11.2 found the same butterfly worth \$0.56, so the trader paid \$0.77 for something worth \$0.56: execution consumed \$0.21 (27% of the price paid, and 38% of the position's theoretical value) before the market moved at all. Section 12.14 shows that this cost recurs in every multi-leg strategy.

12.4 STRADDLES AND STRANGLES: A VIEW ON THE SIZE OF THE MOVE

To bet that something will happen without a view on which way, buy both a call and a put. A **straddle** uses the same strike; a **strangle** uses a lower put strike and a higher call strike, which is cheaper and needs a move at least as large to pay.

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Example 12.4 How far the stock must move

Buy the 47.50 straddle at the asks: call \$2.72 plus put \$2.63 equals \$5.35.

The breakevens are 47.50 ± 5.35, that is \$42.15 and \$52.85. From the current \$48.30 the stock must fall 12.7% or rise 9.4% within 45 days for the position to make a cent.

A strangle using the 45 put (\$1.89 ask) and the 50 call (\$1.51 ask) costs \$3.40, 36% less, but its breakevens are \$41.60 and \$53.40, outside the straddle's on both sides. The strangle costs less and risks less, but it needs a larger move, because the market has priced both structures off the same volatility surface.

This is the general result. An option position's cost and the move it requires are two views of one number, the implied volatility. Buying a straddle is not a bet that the stock will move; the market already expects it to. It is a bet that the stock will move *more than the implied volatility says*—and Chapter 19 documents that implied volatility exceeds subsequent realized volatility on average. The straddle buyer is systematically paying a premium for that bet.

Figure 12.3 draws the two positions of Example 12.4 side by side.

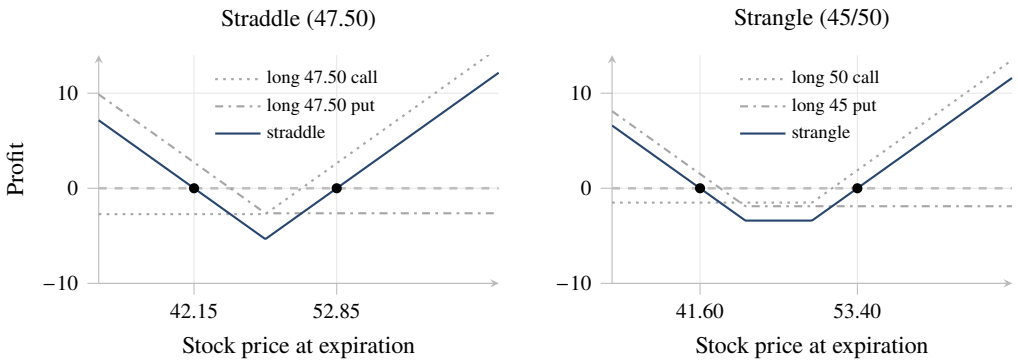


Figure 12.3: Profit at expiration for the straddle and strangle of Example 12.4, at the quotes of Table 10.2, with the legs in gray.

The raw straddle is not a pure volatility trade. An at-the-money straddle has almost no exposure to the direction of a small move, because the call gains about as much as the put loses. That balance holds only while the stock stays near the strike. If the stock rises, the call moves into the money and comes to behave more like the stock, while the put moves out of the money and matters less, so the position as a whole now gains when the stock rises and loses when it falls: it has become net long. Chapter 18 measures that sensitivity as the option's delta and shows why it changes with the stock price. The consequence needs no measurement to see: a straddle held through a large move ends up as a directional position in whichever direction the stock happened to go.

A trader who genuinely wants only volatility exposure must **delta-hedge**: trade the underlying continuously to keep the net directional exposure at zero, so that what remains is a bet on realized against implied volatility alone. Chapters 18 and 22 develop the trade: Chapter 18 measures the gamma (how fast the delta changes as the stock moves) and the theta (the value lost each day to time decay) that the hedge trades against each other, and Section 22.1 shows that the hedging profit accumulated along the way is the difference between realized and implied variance, weighted by gamma. The unhedged straddle of Example 12.4 is a coarser instrument: it contains the volatility bet, plus a directional bet the trader did not intend and is not being paid for.

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12.5 CALENDAR SPREADS AND THE TERM STRUCTURE

A **calendar spread** sells a near-dated option and buys a longer-dated one at the same strike. It is not a directional trade and not straightforwardly a volatility trade; it is a bet on two things at once.

The first is **time decay**. A near-dated option loses time value faster than a long-dated one: Section 11.1 noted that value grows roughly with $\sqrt{T}$, so the decay per day is steepest at the end. Selling the fast-decaying option against the slow one collects that difference, provided the stock stays near the strike. Figure 12.4 shows both halves of the argument: the left panel is the spread's profit at the near expiration, a hump that peaks at the strike and falls away on either side, and the right panel is time value against remaining life, with the near-dated option sitting on the steep part of the curve.

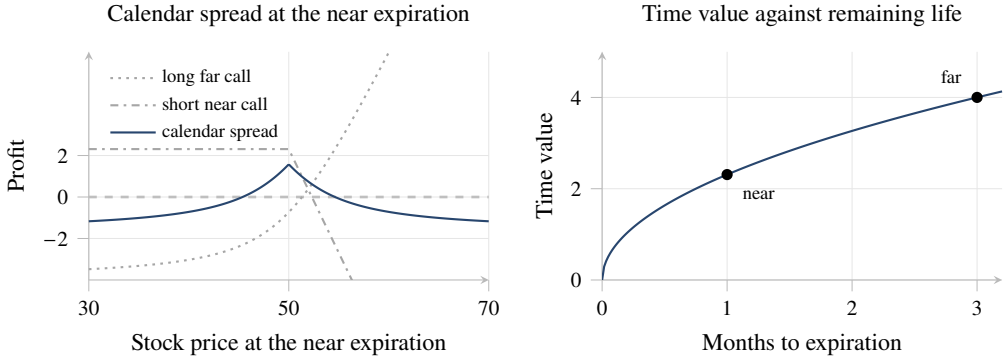


Figure 12.4: Left: profit on a calendar spread at the near option's expiration, $K = 50$, schematic. Right: time value against remaining life, proportional to $\sqrt{T}$.

The second is the **term structure of implied volatility**. The two legs are usually priced at different implied volatilities, and the calendar spread gains if the far option's implied volatility rises and loses if the near one's does. When near-dated implied volatility is elevated (before an earnings announcement, say), selling the near option at that level against a far-dated option priced at a calmer one is the trade the structure is designed for. Chapter 19 treats that term structure properly.

A **diagonal spread** changes the strike as well as the maturity, combining the calendar bet with a directional one. It is popular and frequently misunderstood, because its holder often has a clear view on only one of the two bets they have taken.

12.6 RISK REVERSALS AND RATIO SPREADS: A VIEW ON SKEW

A **risk reversal** buys an out-of-the-money call and sells an out-of-the-money put. Section 11.5 showed that at the same strike this is a synthetic long; at different strikes it is a long position with a gap in the middle, and it can often be established for zero cost.

What makes it interesting is what determines that cost. Buying the call and selling the put trades one implied volatility against another, so the price of a risk reversal *is* the skew: the amount by which downside protection is more expensive than upside participation. In equity markets the lower-strike put almost always carries the higher implied volatility, because puts are persistently in demand, so the trader who buys the call and sells the put is *paid* the skew in volatility terms: the option sold is the dear one and the option bought the cheap one. In dollars the one in Section 12.7 is entered for a small credit, and since its two strikes sit about equally far from the forward, that credit is mostly the skew. Chapter 19 explains why it is there.

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Risk reversals are therefore the standard way to trade a view on skew rather than on the level of volatility. They are also the standard corporate hedge: a firm wanting downside protection at zero premium sells away the upside to pay for it, which is a collar (a put bought and a call sold against a holding; Section 12.9) and is the same structure viewed from the other side. That side pays the skew; Examples 12.5 and 12.6 measure what it costs.

A **ratio spread** buys one option and sells more than one further out. It is cheap to establish, sometimes for a credit, and profits over a range, but the extra short options leave the position exposed beyond them. The shape looks safe on a profit diagram and is not, because the loss is unbounded (for calls; very large for puts) in exactly the tail the trader expects not to occur. Figure 12.5 shows both: the risk reversal is a long position with a flat stretch between the strikes, and the ratio spread's profit turns down without limit beyond the strike it sold twice.

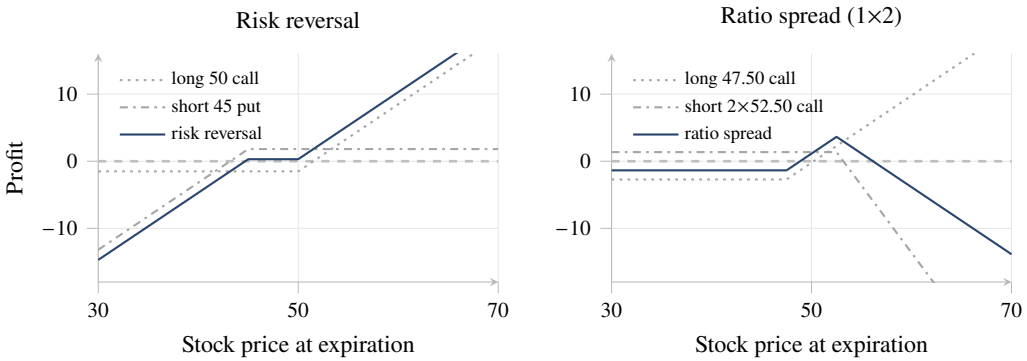


Figure 12.5: Profit at expiration for a risk reversal and a 1×2 ratio spread at the quotes of Table 10.2, with the legs in gray.

12.7 EXPRESSING ONE VIEW FOUR WAYS

The abstract discussion becomes concrete when a single view is priced through several instruments. Suppose a trader believes the stock, at \$48.30, will rise to around \$52 within 45 days. Table 12.1 shows what happens under four ways of acting on that: buying the stock, buying the 47.50 call outright, the 47.50/52.50 bull spread of Section 12.2, and a risk reversal. The outright call is the spread's own long leg, so the second and third columns differ only by the call the spread sold.

The risk reversal here buys the 50 call at \$1.51 and sells the 45 put at \$1.81, for a net credit of \$0.30. Its capital line is not that credit. The written put must be margined, and formula (a) of Section 10.8 applied to it (20% of \$48.30, less the \$3.30 by which the put is out of the money, plus the \$1.81 premium) requires \$8.17 a share, so the route commits $817 - 30 = \$787$. Four things in the table are worth extracting.

The structure tailored to the view wins when the view is exactly right. At \$52, the trader's actual target, the bull spread earns \$246, more than either other option route, and returns 121% on its capital, more than any route; only the stock earns more in dollars, \$370, with twenty-four times the capital. This is what it means for a strategy to match a view.

It loses when the view is right about direction and wrong about size. At \$56 the spread earns \$296 while the outright call earns \$578. The spread sold that outcome away, which is precisely how it was made cheap.

Zero cost is not zero risk. The risk reversal was entered for a \$30 credit and still loses \$270 at \$42, almost as much as the outright call, which cost \$272. Its cheapness comes from the written put, and the written put is the whole downside. A structure that costs nothing has not

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Table 12.1: Profit per 100 shares of exposure, at the quotes of Table 10.2, and the return on the capital each route commits.

Stock at expiration	Buy stock	47.50 call	47.50/52.50 spread	Risk reversal
Profit, $				
$42.00	−630	−272	−204	−270
$45.00	−330	−272	−204	+30
$48.30	0	−192	−124	+30
$52.00	+370	+178	+246	+230
$56.00	+770	+578	+296	+630
Capital committed	$4,830	$272	$204	$787
Return on capital committed				
$42.00	−13.0%	−100%	−100%	−34.3%
$45.00	−6.8%	−100%	−100%	+3.8%
$48.30	0	−70.6%	−60.8%	+3.8%
$52.00	+7.7%	+65.4%	+120.6%	+29.2%
$56.00	+15.9%	+212.5%	+145.1%	+80.1%

eliminated risk; it has financed one exposure by selling another.

Capital efficiency and risk run in opposite directions. Buying the stock ties up $4,830 and loses $630 in the bad case; the spread ties up $204 and loses $204. The return block makes the leverage explicit: at the target the spread returns 121% on its capital and the stock 7.7%, and at $42 the same spread returns −100% while the stock loses 13%. A return on capital is meaningless unless the loss column is read at the same time, which is the point Example 1.5 made about leverage, appearing here as a choice among strategies.

12.8 BUILDING AN ARBITRARY PAYOFF

The structures above are conventional, but nothing restricts a trader to them. A general result underlies the whole subject and is worth stating explicitly, because it explains where exotic payoffs come from and is the bridge to Chapters 19, 22 and 34.

Any piecewise-linear payoff can be replicated exactly by a bond, a forward, and a portfolio of European calls. The recipe is mechanical: place a call at each kink, with a quantity equal to the *change in slope* at that kink.

Take the bull call spread. Its payoff is flat at zero below $47.50, rises with slope 1 to $52.50, and is flat above. The slope changes by +1 at 47.50 and by −1 at 52.50, so the replicating portfolio is one long call at 47.50 and one short call at 52.50—which is what a bull spread is. The butterfly's slopes are 0, +1, −1, 0, giving changes of +1, −2, +1: long one, short two, long one.

The same recipe builds anything. A note paying 70% of an index's rise, capped once the index has risen 30% (a maximum return of 21%), is a payoff with slope changes of +0.7 at the initial level and −0.7 at 130% of it: buy 0.7 calls at the first strike, sell 0.7 at the second. Figure 12.6 shows the construction.

Digitals as the limit of a spread. A **digital** (or binary) option pays a fixed amount if the underlying finishes above a strike and nothing otherwise. Its payoff has a jump rather than a kink, so no finite portfolio of calls reproduces it, but a tight call spread approaches it. Buying a call at $K - \varepsilon$ and selling one at $K + \varepsilon$, scaled by $1/(2\varepsilon)$, gives a payoff that rises from 0 to 1 across a narrow band. As $\varepsilon \to 0$ the position becomes a digital, and its price becomes

$$\text{digital}(K) = -\frac{\partial c}{\partial K}$$

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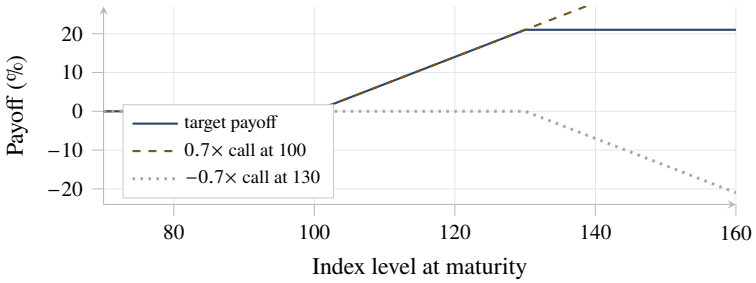


Figure 12.6: A capped participation payoff as two call positions.

This is not merely a pricing device. It is how digitals are hedged: a dealer short a digital holds a tight call spread, and the tighter the spread the larger the position required, which is why digitals are risky to hedge near the strike at expiration: the same pin risk of Example 10.6 in a more acute form. The autocallable coupons of Section 12.12 are digitals, which is why those products are difficult to hedge precisely when the index sits near a barrier. Chapter 19 pushes the same construction one derivative further, to a butterfly of vanishing width, and reads the market's probability distribution for the underlying off the option chain.

12.9 COLLARS AND THE CONCENTRATED SHAREHOLDER

The most common institutional use of an option combination is not speculative. A founder, executive, or family office holding a large position in a single stock wants to reduce the risk without selling, because selling would trigger tax, breach a lockup (an agreement not to sell for a set period, typical after a public offering), or signal something unwanted.

The **collar** buys a put and sells a call to pay for it. Section 12.6 noted it is a risk reversal seen from the other side; here it is a hedge rather than a view.

Example 12.5 A collar, and why its strikes are not symmetric

A shareholder holds 200,000 shares at \$48.30, worth \$9.66 million. Using Table 10.2, they buy the 45 put at its \$1.89 ask and sell the 50 call at its \$1.44 bid:

$$\text{net} = 1.44 - 1.89 = -\$0.45 \text{ per share, a debit of } \$90{,}000$$

The call pays for three-quarters of the put. Leaving that \$90,000 aside, the position is floored at \$45 and capped at \$50:

$$\text{worst case} = 200{,}000 \times 45 = \$9.0\text{m}, \qquad \text{best case} = 200{,}000 \times 50 = \$10.0\text{m}$$

The put is 6.8% below spot; the call is only 3.5% above it. The holder gave up 3.5% of upside to protect against everything below 6.8% down, and still paid.

Two things impose that asymmetry. The first is the forward: the dividend Example 10.3 found in the chain puts the 45-day forward near \$47.60, from which the strikes are nearly symmetric, 5.5% below and 5.1% above, so on this chain the forward sets most of the gap in the strikes. The second is skew: measured from the forward the two options are nearly equidistant, yet the put costs \$0.38 more at mid (\$0.45 at the quotes), because the chain prices it at a 46% implied volatility against the call's 37% (Section 15.11). A zero-cost collar must therefore set its put further from the forward than its call, as Example 12.6 shows on index quotes. Chapter 19 explains why, and the reason connects directly to who is on each side: shareholders like this one persistently buy downside protection, and someone must be paid to supply it.

A practical corollary follows. Because skew is part of what sets the terms, a collar is cheapest to establish when skew is flat.

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12.10 HEDGING A PORTFOLIO WITH INDEX OPTIONS

Section 3.6 hedged an equity portfolio with index futures. Options offer an alternative with a different shape, and the choice between them is the clearest institutional application of everything in this chapter.

The contract count is the same beta-adjusted calculation as for futures:

$$N = \beta \frac{V_P}{S \times \text{multiplier}} \tag{12.1}$$

where V_P is the value of the portfolio, β its beta against the index, S the index level, and $S \times$ multiplier the value of the index underlying one option contract (the role V_F played for a futures contract in Section 3.6). What differs is that a put costs money but can lose no more than that, while a futures hedge is free to establish and removes the upside along with the downside.

Example 12.6 What protecting a portfolio actually costs

A fund holds \$50 million of equities with a beta of 1.1 against an index at 5,250. The index option multiplier is \$100. From (12.1),

$$N = 1.1 \times \frac{50{,}000{,}000}{5{,}250 \times 100} = 1.1 \times 95.24 = 104.8 \rightarrow 105 \text{ contracts}$$

Buying three-month puts 5% out of the money, struck at 4,987.5 and quoted at 60.00 points:

$$\text{cost} = 60 \times 100 \times 105 = \$630{,}000 = 1.26\% \text{ of the portfolio, per quarter}$$

The floor is set on the index, not on the portfolio. With a beta of 1.1 a 5% fall in the index is roughly a 5.5% fall in the portfolio, and the beta-adjusted contract count is what makes the puts pay at that rate. The beta scaling is an approximation that ignores interest and dividends; Section 16.2 derives the strike that corrects for them. If the index falls 10%, to 4,725, the portfolio loses about \$5.5 million while the 105 puts pay $262.5 \times 100 \times 105 = \$2.76$ million, so the loss net of the put payoff is \$2.74 million, or 5.5% of the portfolio, 6.7% once the \$630,000 premium is counted. The floor stays near 5.5% however far the index falls beyond the strike, drifting only with the rounding of 104.8 contracts to 105.

Renewed every quarter, that is roughly 5% a year. Set that against a long-run expected equity return in the high single digits and the conclusion is uncomfortable but unavoidable: continuous protection at a strike this close costs most of the expected return. This single number explains why most institutional hedging structures look the way they do.

The put-spread collar is the usual response. Buy the 4,987.5 put at 60, sell a 4,462.5 put (15% down) at 22, and sell a 5,512.5 call (5% up) at 48:

$$\text{net} = -60 + 22 + 48 = +10 \text{ points, a credit of } \$105{,}000$$

Look first at the two 5% strikes. The put costs 60 points and the call 48. Over three months the index forward sits only about 1% above spot (a 5% interest rate less a dividend yield near 1.5%), which on its own would make the call the dearer of the two, so the gap comes from skew and if anything understates it. This is the same skew Example 12.5 found on a single stock, here with the forward slightly above spot rather than pulled below it by a dividend. A zero-cost collar on this index has to set its call nearer to spot than its put.

The fund is now protected between −5% and −15% on the index (about −5.5% to −16.5% on the portfolio) and capped at +5% on the index, about +5.5% on the portfolio; and it has been paid to enter the structure. What it has given up is protection *below* −15%—precisely the scenario the hedge was ostensibly for. Selling the deep put is part of what pays for the structure, so a budgeted program keeps protection against ordinary declines and gives up protection against catastrophes.

With the bought put moved to the money, wrapped in an exchange-traded fund and reset once a year, this structure is the **buffer ETF**, or defined-outcome fund: over each period the fund absorbs the first 10 or 15% of the index's loss and caps its gain, and such funds held about \$78 billion at the end of 2025. Financial advisers buy them for clients near retirement who want to stay in

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equities with the first losses absorbed, and who accept the cap, and the losses beyond the buffer, in exchange.

Tail hedging inverts the choice: buy puts 30% out of the money on the index (a fall of about 33% in the portfolio) at, say, 6 points, costing $63,000 a quarter, or about 0.5% a year. The premium is small and the payoff enormous when it comes. The difficulty is that Chapter 19 shows deep out-of-the-money puts carry the *highest* implied volatilities of any options on the surface, so it is the dearest insurance relative to its expected payoff, although its premium is the smallest. A strategy costing 0.5% a year that pays off only in a rare crash must pay a great deal on those occasions to justify itself, and whether it does is genuinely contested.

As Section 3.2's case study warned, an option premium sits in a named budget line every year it pays nothing, which is part of why collars, which can be struck at no cost, are popular.

Options instead of futures: the margin asymmetry

A commercial hedger faces the same choice, and there the decisive difference is not the premium at all. A futures hedge fixes the price in *both* directions and demands cash on every day the market moves against it. A purchased option (a producer's put, a consumer's call) whose premium is paid up front removes only the unfavorable tail, and owes no variation margin (unless the option itself is margined futures-style, Section 16.5), because the worst outcome for a purchased option is that it expires worthless.

That asymmetry is what Section 3.10's hedging failures are made of. Every one of them runs through variation margin on a futures leg, and a hedger holding purchased options instead—puts for a producer, calls for a seller of fixed-price supply like Metallgesellschaft—would have faced no margin call in any of them. The premium therefore buys two distinct things: the retained upside, and the elimination of funding risk. Whether the second is worth paying for depends on the firm's access to liquidity, which is exactly the judgment Section 3.10 says firms get wrong.

Writing options is a different matter entirely. A firm that sells options to reduce its hedging cost (selling calls to fund puts, the collar above) takes the margin obligation back on the written leg, along with an exposure on that leg that is not capped. The premium saved is not free, and such a structure's cash flows can come to resemble the futures hedge it was meant to improve on.

12.11 WHAT EACH STRATEGY IS ACTUALLY A VIEW ON

Table 12.2 is the chapter in one place. It records what each structure bets on, what it costs, and what it can lose.

Three observations organize the table.

The time column is almost the volatility column reversed. Positions that are short volatility are helped by the passage of time, and positions that are long volatility are hurt by it. This is not a coincidence: Chapter 18 shows that gamma and theta are tied together: a position cannot hold gamma without paying theta, or sell it without collecting theta. The practical meaning is immediate: you are paid to wait or you pay to wait, and which one is determined by whether you are net long or short options.

Almost nothing is a pure directional bet. Only the outright option, the **vertical spreads** of Section 12.2 (spreads whose legs share an expiration and differ only in strike), and the risk reversal have direction as their dominant exposure. Everything else requires the trader to be right about magnitude or timing as well, and a strategy chosen for its payoff shape frequently embeds a volatility view its holder never formed.

The neutral strategies are all short a sharp move. Butterflies, condors, and calendars all profit from stillness, which means they all lose in a sharp move, even the calendar, which Table 12.2 lists as "term" because its far-dated option gains when implied volatility rises. A

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Table 12.2: Exposures of the main structures, with the net premium and the maximum loss per share at the quotes of Table 10.2 (the covered call writes the 50 call; the protective put buys the 45 put).

Strategy	Direction	Volatility	Time	Net premium	Maximum loss
Covered call	mildly up	short	+	credit 1.44	46.86
Protective put	up	long	−	debit 1.89	5.19
Bull call spread	up	mixed	mixed	debit 2.04	2.04
Bear put spread	down	mixed	mixed	debit 2.14	2.14
Butterfly	neutral	short	+	debit 0.77	0.77
Condor	neutral	short	+	debit 1.70	1.70
Straddle	none	long	−	debit 5.35	5.35
Strangle	none	long	−	debit 3.40	3.40
Calendar spread	neutral	term	+	debit	the debit
Risk reversal	up	skew	mixed	credit 0.30	44.70
Ratio spread	mildly up	short	+	debit 1.36	unlimited

portfolio of "diversified" neutral strategies is a concentrated short-volatility position, and it will be discovered to be one on the day it matters.

The table also says whose book each structure ends up on. Vertical spreads belong to directional traders who want a view at a known maximum loss, which is also what the position is margined at (Section 12.14). Straddles and strangles are bought around earnings and central-bank meetings by event traders, and delta-hedged by volatility funds; market makers are usually on the other side. Butterflies and condors, which profit from stillness, suit traders who expect a quiet market, and premium-selling funds run condors in the iron form of Section 12.3. Calendars and diagonals are used by term-structure traders and by dealers managing the maturity profile of their volatility exposure, selling options in the nearest expiration at elevated implied volatility against later ones. Ratio spreads are mostly professional: a dealer or hedge fund uses one to reduce the cost of a directional view, and a put ratio spread sells the low-strike puts whose implied volatility the skew has raised.

12.12 STRUCTURED PRODUCTS ARE PACKAGED STRATEGIES

Most people who own an option strategy did not choose it. They bought a product, and the product is a strategy with a wrapper, a fee, and a name.

A principal-protected note promises the return of capital plus participation in an index. It is a zero-coupon bond plus a call.

A reverse convertible pays a high coupon and repays principal in shares if the stock finishes below its initial level (in the common barrier form, only if it has also traded through a lower barrier). It is a bond plus a written put: the high coupon is the put premium, as Example 12.8 shows.

An autocallable pays a coupon and redeems early if the index is at or above a level on an observation date, and can return less than principal if it breaches a lower one. It is a written **down-and-in put** (a put that comes into existence only if the index falls through the lower level, the barrier) plus a series of digital coupons. The Chinese "snowball" products of Section 35.7 are exactly this structure.

Example 12.7 Taking apart a principal-protected note

A five-year note returns 100% of principal at maturity plus α times any rise in an index. What can α be?

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The issuer must set aside enough to repay principal. At a five-year rate of 4.5%,

$$\text{zero-coupon bond} = 100\,e^{-0.045\times 5} = 100\times 0.798516 = \$79.85$$

leaving \$20.15 per \$100 to buy call options. If a five-year at-the-money index call costs 25% of notional,

$$\alpha = \frac{20.15}{25} = 80.6\%$$

Subtract a 2% distribution fee and the budget falls to \$18.15, giving $\alpha = 72.6\%$. The 2% fee cost the investor 8 percentage points of participation, because it is taken out of the \$20.15 option budget, a tenth of it, rather than out of the whole note.

Now vary the rate, holding the call's cost at 25%. At a 1% rate the bond costs $100e^{-0.05} =$ \$95.12, leaving \$4.88, and after the fee \$2.88, for participation of 11.5%. At a 5% rate the bond costs \$77.88, leaving \$22.12, and after the fee \$20.12, for participation of 80.5%.

This explains an entire product cycle. Principal-protected notes were nearly impossible to build attractively through the years of near-zero interest rates, and higher interest rates make them attractive to build again. No change in investor preferences is needed to explain it. The option budget is what is left after the bond, and when the bond is expensive there is nothing left.

Example 12.8 Where a 12% coupon comes from

A one-year reverse convertible on a stock at \$100 pays a 12% coupon. If the stock finishes at or above \$100 the investor receives \$100; if below, they receive the shares instead. The one-year rate is 4.5% and a one-year at-the-money put on the stock is worth \$9.00.

Decompose what the investor owns. They hold a bond, they receive the coupon, and they have written a put struck at \$100:

$$\begin{aligned}\text{bond} &= 100\,e^{-0.045} = \$95.60\\ \text{coupon} &= 12\,e^{-0.045} = \$11.47\\ \text{written put} &= -\$9.00\\ \text{total} &= \$98.07\end{aligned}$$

The investor pays \$100 for a package worth \$98.07. The \$1.93 difference is the fee, and it is not disclosed as one.

More important is where the 12% comes from. It is not compensation for credit risk and has nothing to do with the issuer's creditworthiness beyond the bond leg. Of the coupon, roughly 4.5 points are the risk-free rate and the remainder is the premium on a put the investor has sold, less the fee just computed. The high coupon is insurance income, and the risk being insured is exactly the risk the investor is taking.

The product is therefore attractive in the same conditions as the covered call of Example 10.7, and unattractive in the same ones. It performs worst when the stock falls sharply, which is when the investor most wanted the \$100 back and instead receives shares worth considerably less. A useful test for any high-coupon structured product: work out what option the buyer has written, and ask whether they would have written it deliberately.

Case Study 12.1 The covered-call fund boom

Funds that systematically write calls against an equity portfolio (sold as "derivative income" or "buy-write" strategies) grew from a niche into one of the fastest-gathering categories in the ETF market during 2022–2024, with assets approaching \$100 billion by the end of 2024.

The appeal is a headline distribution yield far above anything available in equities or, for much of the period, in bonds. The mechanics of Section 12.1 explain where it comes

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from: the fund is short puts, and the yield is insurance premium. Chapter 19 shows the premium is on average positive, so this is a real source of return and not an illusion.

Two things are nevertheless misdescribed. First, the distribution is not income in the sense investors usually mean. It is partly the option premium and partly a return of the investor's own capital, and a fund can keep paying a high distribution while its net asset value falls, because whatever the premium and dividends do not cover is paid out of the fund's own assets. Second, and more important, the strategy gives up the strong markets (Example 10.7). A fund that caps the upside every month will, over a long horizon containing a few large rallies, badly trail the index it writes against—while having delivered exactly the yield it advertised throughout.

The fund has no maturity and promises nothing about what an investor ends up with: it writes new calls every month indefinitely, and its net asset value floats with the index, less the upside it sells and the capital it pays out. These products convert an uncertain total return into a higher, steadier distribution and a lower expected terminal value. For an investor who needs cash flow and can accept that trade, they do something useful. For an investor who reads the distribution yield as income without cost, they are a slow disappointment, and the disappointment arrives only after several years.

12.13 MANAGING A POSITION AFTER ENTRY

Every profit diagram in this chapter describes a position held to expiration. Most are not, and the decisions taken in between determine the outcome at least as much as the structure chosen at entry.

Rolling. A covered call writer whose stock has risen through the strike faces assignment and the loss of the shares. The standard response is to **roll**: buy back the written call and sell another at a higher strike or a later date. Usually this costs money, since the call being repurchased is now in the money.

Rolling once is ordinary position management. Rolling repeatedly is something else. Each roll realizes a loss on the option leg and re-establishes a short call further out, so a writer who rolls to avoid assignment has converted a defined, capped strategy into an open-ended bet that the stock will stop rising—the one thing they have so far been wrong about. The strategy that was chosen for its conservatism becomes, through a sequence of individually reasonable adjustments, a persistent short call on a rising stock.

Repairing. A trader holding a long call on which the stock has fallen can sell a higher-strike call against it, converting the position into a spread and recovering part of the premium. The arithmetic is unkind: the stock has fallen, so the higher-strike call is now cheap, and the repair raises very little.

Pre-committing to exits. A stock position can be held indefinitely; an option position cannot. Time decay means "wait and see" is a decision with a price, and Table 12.2 gives its sign for each structure. The discipline that follows is to decide before entry what will cause the position to be closed (a target, a stop, or a date), because after entry the decay makes every delay look like a reason to keep waiting for the move that has not yet come.

Adjustments accumulate costs. Each adjustment adds legs, and Section 12.14 shows what legs cost. A butterfly adjusted twice has crossed eight more bid–ask spreads on top of the four at entry, and on the numbers of Example 12.3 the twelve together exceed the position's entire

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theoretical value. Frequent adjustment is not risk management; past a point it is a transfer to the market maker.

12.14 THE COSTS THAT DECIDE WHETHER A STRATEGY WORKS

Every profit diagram in this chapter is drawn as though the position could be established at a single price. It cannot, and the gap between the diagram and the outcome is usually decisive.

Each leg pays the bid–ask spread. Example 12.3's butterfly lost 38% of its value at entry, and a four-legged condor crosses four spreads. The number of legs multiplies the transaction cost, and the structures with the most attractive reward-to-risk ratios usually have the most legs.

Exit costs the same again. A butterfly held to expiration avoids the exit spread but accepts whatever the stock does. Closing early pays the spread a second time, which on the numbers above would consume three-quarters of the theoretical value.

Legging risk is real. The risk of Section 10.5, that the market moves between fills, is larger here: on a \$0.56 butterfly built from options worth \$4.425, \$2.67, and \$1.475, a move of about 50 cents in the underlying between fills (1% of the stock price) can exceed the entire position value.

Margin treatment varies. A spread whose loss is capped requires margin equal to that maximum loss; a ratio spread with uncovered short options requires the naked-writing margin of Section 10.8. Two structures with similar diagrams can have very different capital requirements, and the more attractive-looking one is often the more expensive to hold.

Short American legs can be assigned. The case study in Section 11.7 showed a supposedly riskless box destroyed by early assignment. A short call is most likely to be exercised just before an ex-dividend date, when the dividend exceeds its remaining time value (Example 11.7); a short put, once it is deep in the money (Section 11.9). Either way one leg disappears at a moment chosen by someone else: a bull call spread whose short call is assigned becomes a long call against short stock, a synthetic long put.

Put together, these determine which strategies are practical for whom. A market maker with negligible transaction costs, the portfolio margining of Section 10.8 (margin set on the net risk of the whole account rather than leg by leg), and the ability to trade packages can hold structures that are not practical for a retail trader crossing full spreads on every leg. The correct strategy depends on who is trading it—a question textbooks usually leave implied.

SUMMARY

Every option position is a bet on some combination of direction, magnitude, time, and skew. Choosing a strategy means matching those exposures to a view actually held.

Put–call parity collapses much of the apparent variety. A covered call is a written put plus cash; a protective put is a purchased call plus cash. Where two routes to one exposure exist, the one with fewer option legs is usually cheaper to trade.

Spreads cap the payoff in exchange for a lower cost and are bets on a range rather than on a direction. Butterflies and condors bet on stillness. Straddles and strangles bet on magnitude—but because both the cost and the required move come from the same implied volatility, a straddle is a bet that realized volatility will exceed implied, not that the stock will move.

Calendar spreads bet on time decay differentials and on the term structure of volatility. Risk reversals bet on skew, and are the same structure as the corporate collar.

Strategies that are neutral on direction are almost all short volatility, so a portfolio of them is concentrated rather than diversified.

Any piecewise-linear payoff can be replicated by a bond, a forward, and calls, with weights equal to the change in slope at each kink. A digital is the limit of a tight spread, priced at $-\partial c/\partial K$, and that is how dealers hedge it.

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Structured products are packaged strategies: a principal-protected note is a bond plus a call, a reverse convertible is a bond minus a put, an autocallable is a written barrier put plus digital coupons. Because the option budget is whatever remains after buying the bond, participation rates depend sharply on the level of interest rates, which is why principal-protected notes became scarce in the years of near-zero interest rates.

Transaction costs decide which strategies are practical. Paying the bid–ask spread on every leg can consume a large share of a butterfly's mid-price value, and structures with the best reward-to-risk ratios generally have the most legs and therefore the highest costs. Margin treatment, legging risk, and early assignment on American legs complete the picture, and together they mean the right strategy depends on who is trading it.

FURTHER READING

The OCC's *Characteristics and Risks of Standardized Options* and McMillan's *Options as a Strategic Investment* set out the standard structures and their margin treatment, which changes more often than textbooks do. On covered calls, Whaley (2002) built the buy-write index and reports its first long-horizon record, and Israelov and Nielsen (2015) decompose the strategy into the equity, short-volatility and timing exposures this chapter identifies. Henderson and Pearson (2011) take a retail structured note apart into a bond and options, as Example 12.8 does, and find it priced about 8% above its value; it is the model for doing the exercise on any term sheet. Carr and Madan (2001) state the replication result of Section 12.8 in general, and Breeden and Litzenberger (1978) take it to the butterfly of vanishing width and the risk-neutral density.

PRACTICE QUESTIONS

12.1 Show, using put–call parity, that a covered call has the same profit profile as a written put. State one respect in which the two positions nevertheless differ in practice.

12.2 Using Table 10.2, construct a bear put spread from the 47.50 and 45.00 puts at the quoted bid and ask. Compute the debit, the maximum profit, the maximum loss, and the breakeven.

12.3 Using Table 10.2, price the 47.50/50/52.50 call butterfly at the quoted bid and ask and at mid prices. Compute what execution costs as a percentage of the mid value, both breakevens at the quotes, and where the profitable range lies relative to the \$48.30 stock price.

12.4 Explain why a strangle that costs 36% less than a straddle can require a larger move to break even. What does this imply about "cheap" option strategies generally?

12.5 A five-year principal-protected note is offered when the five-year rate is 3% and a five-year at-the-money call costs 22% of notional. Compute the participation rate with no fee and with a 1.5% fee.

12.6 Identify what each of the following is a bet on, using the four things to bet on named in the chapter's introduction: (a) a calendar spread; (b) a risk reversal; (c) a condor; (d) a covered call.

12.7 Explain why a portfolio containing butterflies, condors, and calendar spreads is not diversified.

12.8 A trader establishes a four-legged condor by executing each leg separately. Describe two distinct ways this can cost more than expected, and state how each is avoided.

12.9 A calendar spread sells the near-dated option and buys the longer-dated one at the same

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strike. Explain why this is neither a directional trade nor straightforwardly a volatility trade, identify the two effects that determine its profit, and state the move in the underlying that hurts it most.

12.10 A risk reversal buys an out-of-the-money call and sells an out-of-the-money put. Explain why its cost is a direct reading of the skew rather than of the level of volatility, and state what a risk reversal that costs nothing to enter tells you about the two implied volatilities.

12.11 A founder holds a large single-stock position and wants to reduce the risk without selling. Describe the collar that achieves this, explain in whose favor the skew works when the collar is on an equity, and state what the shareholder remains exposed to that a sale would have removed.

12.12 A manager hedges an equity portfolio with index puts rather than index futures. State the beta-adjusted contract count, explain what shape the option hedge buys that the futures hedge does not, and explain why the floor it sets on the portfolio is only approximate even when the beta is correct.

12.13 A producer can hedge by selling futures or by buying puts. Compare the two on cost at inception, retained upside, and cash demands over the life of the hedge, and identify which consideration the textbook comparison usually omits.

FURTHER QUESTIONS

12.14 Example 12.3 found that execution consumed 38% of a butterfly's theoretical value. Derive a general expression for the fraction of a multi-leg strategy's value consumed by spreads, in terms of the number of legs and the proportional spread on each. Then use it to explain why market makers can trade structures retail traders cannot.

12.15 A fund writes one-month at-the-money calls against an equity index every month, collecting a premium of 1.5% of the index each time. Construct the sequence of monthly outcomes over a period containing eleven months of +1% and one month of +15%, and compare the fund's total return with the index's. Then explain why a backtest beginning in a calm period will overstate the strategy's appeal.

12.16 Section 12.12 decomposes an autocallable into a written barrier put plus digital coupons. Using that decomposition, explain what happens to the issuer's hedge as the index approaches the knock-in barrier, and connect your answer to the 2024 episode described in Section 35.7.

12.17 A corporate treasurer is offered a zero-cost collar on a large shareholding. Identify what the treasurer is selling, what determines whether the collar can genuinely be struck at zero cost, and under what market conditions the terms will be most attractive.

12.18 Table 12.2 lists the calendar spread as "term" rather than long or short volatility. Explain precisely what exposure this means, construct a case in which implied volatility rises across all maturities and the calendar spread nevertheless loses money, and state what the position is really long.

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